

PPOM-ATTACK: A Substitute Model-Free Perturbation Prediction and Optimization Method for Black-Box Adversarial Attack Against Face Recognition

Ke Cheng[✉], Jixin Zhang[✉], *Member, IEEE*, Haiyun Li[✉], Zipeng Zhong[✉], Mingwu Zhang[✉],
and Zheng Qin[✉], *Member, IEEE*

Abstract—Face recognition (FR) brings convenience to people’s lives while also posing security risks. Some malicious users employ FR attacks to impersonate the identity of a target. To reveal the security risks, recent work has attacked black-box FR models by utilizing substitute models to generate adversarial face images that are misclassified as the target individual due to the attack transferability of substitute models. However, the substitute models cannot accurately approximate the target model that leads to a decrease in FR attack success rate and adversarial face image quality. To address the issue, we propose the PPOM-ATTACK, a substitute model-free Perturbation Prediction and Optimization Method for black-box adversarial Attack against face recognition. PPOM-ATTACK directly obtains feedback from the target model instead of using substitute models, it avoids any discrepancy with the attack objective. To achieve this goal, we design a proximal policy optimization (PPO)-based agent to predict the perturbation regions in the face image and self-adaptively disturb the regions. To maintain high-quality adversarial face images, we further propose a minimum brightness offsets method specifically designed to generate perturbations that minimize the feature embedding difference between the adversarial and targeted face images. The experimental results show that our approach outperforms state-of-the-art FR attack methods by an average of 21.7% in terms of attack success rate, while achieving better image quality on seven FR models.

Index Terms—Adversarial attack, deep reinforcement learning, face attack, face recognition.

I. INTRODUCTION

FACE Recognition (FR) techniques are widely used in our daily life, such as electronic payment system, access control system, etc. However, due to the vulnerability of the artificial intelligence algorithms, FR models

pose privacy and security risks. Malicious users can slightly disturb one’s face image with invisible changes as the adversarial face image, but make FR models identify the adversarial face image of the original person as the target person.

To reveal the security risks in face recognition, recently, adversarial face image attack becomes one of the most hot topics in artificial intelligence security. Recent works, such as [1], [2], [3], [4], [5], [6], [7], [8], [9], and [10], usually use substitute models instead of the target FR models since the gradients of the target FR models are always unknown (black-boxed) in reality, and use generative models such as gradient-based models, Generative Adversarial Network (GAN), Stable Diffusion (SD), etc. to generate adversarial face images according to the gradient information from the substitution model and the original face images. For example, criminals may use some existed FR models as the substitute models to fit the target black-box FR model. The substitute FR model will be treated as the pre-trained discriminative model. The discriminative model and generative model are two parts of GAN. Malicious users train a generative model through adversarial training with the discriminative model and similarity maintaining training with the original face image. The converged generative model generates the adversarial face image which is similar to the original face image. Due to the assumption of transferability of the substitute model, the adversarial face image can not only fool the substitute FR model but also fool the target FR model.

However, the assumption of transferability of the substitute model is not always true. Since the architectures and training data of the substitute FR models and the target FR models are always quite different, the substitute FR models cannot accurately approximate the target FR model, so that the adversarial face image can only successfully attack the substitute FR model, but cannot efficiently attack the target FR model. When using the substitute FR model, there is a significant decrease in FR attack success rate and adversarial face image quality, as shown in Figure 1.

To address the above issue, in this paper, we propose a substitute model-free method that directly obtains feedback from the target black-box FR model instead of using substitute FR models, it avoids any discrepancy with the attack objective. To achieve this goal, two main technical challenges must be addressed:

Received 15 June 2025; revised 1 December 2025 and 27 January 2026; accepted 13 February 2026. Date of publication 23 February 2026; date of current version 12 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62441233, Grant 62002106, Grant 62472150, Grant 62072134, Grant 62472166, and Grant U22A2030; and in part by the Major Research Plan of Hubei Province under Grant 2023BAA027. The associate editor coordinating the review of this article and approving it for publication was Prof. Annalisa Franco. (Corresponding author: Jixin Zhang.)

Ke Cheng, Jixin Zhang, Zipeng Zhong, and Mingwu Zhang are with the School of Computer Science, Hubei University of Technology, Wuhan 430068, China (e-mail: zhangjx@hbut.edu.cn).

Haiyun Li is with Shenzhen International Graduate School, Tsinghua University, Shenzhen 518071, China.

Zheng Qin is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China.

This article has supplementary downloadable material available at <https://doi.org/10.1109/TIFS.2026.3666853>, provided by the authors.

Digital Object Identifier 10.1109/TIFS.2026.3666853

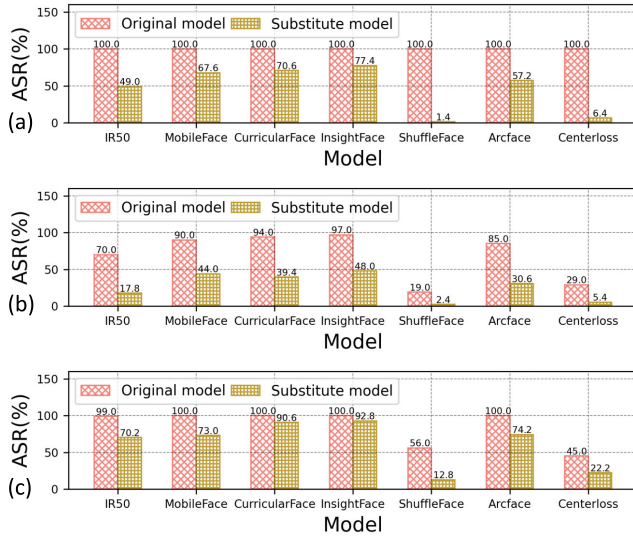


Fig. 1. Attack success rates of several state-of-the-art methods targeting both the original (target) FR model and substitute FR models are as follows. (a) Sibling-Attack, where the substitute FR model is the multi-task IR152. (b) Clip2Protect, where the substitute FR model is an ensemble of remaining models except the target FR model. (c) AdvDiffusion, where the substitute FR model is an ensemble of models except the target FR model.

- One of the biggest challenges is that the feedback from the target FR model contains little gradient information which leads to inefficient generation of adversarial face image, even random perturbation, due to the unknown perturbation location.
- Without the gradient information of the substitute FR model, the perturbations added to the face image may contains more noise, which leads to a decrease of adversarial image quality.

To address the challenge of non-gradient information which results in unknown perturbation location of face images, we need to predict the perturbation region instead of using gradient information to perturb face images. To address the challenge of decrease of adversarial image quality without gradient information, we need to minimize the perturbation magnitude on face images. In this paper, we propose a substitute model-free perturbation prediction and optimization method for black-box adversarial attack against FR models. Specifically, we first design a proximal policy optimization (PPO)-based agent to predict the perturbation regions in the face image and self-adaptively disturbing the regions. The PPO-based agent adapts its perturbation strategy by learning the impact of perturbation regions through feedback from the target FR model, and makes a decision on which region to disturb in order to improve the attack efficiency. We then propose a minimum brightness offsets optimization method to maintain the high-quality of adversarial face images. It is specifically designed to generate perturbations that slightly change the brightness of the face region in order to minimize the feature embedding difference between the adversarial and targeted face images.

The main contributions are summarized as follows:

- We propose the PPOM-ATTACK, a substitute model-free perturbation prediction and optimization method for

black-box adversarial attack against FR models, which directly obtains feedback from the target black-box FR model instead of using substitute FR models, to avoid any discrepancy with the attack objective.

- We propose a PPO-based agent to predict the perturbation regions in the face image and self-adaptively disturbing the regions to achieve efficient adversarial face image attack, and propose a minimum brightness offsets method to maintain the high-quality of adversarial face images.
- The experimental results on three different datasets show that our approach achieves 21.7% higher average Attack Success Rate (ASR) and the better image quality compared with the state-of-the-art methods across various open-sourced and commercial FR models. Specifically, against adversarially defended models, our method exceeds 15.8% of average ASR compared with the state-of-the-art methods. It demonstrates that the substitute model-free approach can achieve high efficiency of FR attack without gradient information.

II. RELATED WORKS

A. Adversarial Attacks Against FR Models

Adversarial attacks against FR models raise significant concern in machine learning due to their potential impact on security and safety-critical applications [11], [12], [13], [14], [15], [16]. Substitute model-based attacks use generative models such as gradient-based models, Generative Adversarial Network (GAN), Stable Diffusion (SD), etc. to generate adversarial face images according to the gradient information from the substitution model and the original face images. Query-based adversarial attacks rely instead on interaction with the target model through queries to obtain prediction labels or confidence scores and optimize adversarial perturbations by strategically probing the model's decision boundaries.

1) *Substitute Model-Based Adversarial Attacks*: Recently, several approaches have been proposed to enhance the transferability of adversarial attacks by designing underlying gradient-based algorithms such as PGD, TI-RDIM, etc. [1], [2], [3], [17], [18], [19], [20], [21], [22]. In PGD [1], the authors iteratively apply projected gradient updates to generate stronger adversarial examples under the constraint. In TI-RDIM [2], the authors improve transferability by incorporating translation-invariant gradients to better align surrogate and target decision boundaries. In TIP-IM [3], the authors generate targeted adversarial identity masks that can be overlaid on face images for identity protection. In Sibling-Attack [17], the authors leveraged extra information from multi-correlated tasks to boost attacking transferability. In ATT [22], the authors boost adversarial transferability by randomly skipping transformer encoder blocks to induce attention uncertainty and diversify attack trajectories. In Diverse Parameters Augmentation (DPA) [21], the authors enhance adversarial transferability by constructing a surrogate ensemble with diverse parameter initializations and refining perturbations via feature-level aggregation. Although these gradient-based methods can generate precise perturbations according to the substitute model gradients, their tendency to over-fit to the substitute model

used for gradient computation results in a decrease in attack success rates due to differences between the substitute model and the target model.

Regarding the GAN-based adversarial attacks against the FR task, In the works of Adv-Makeup [8], AMT-GAN [4], Adv-Attribute [23], Clip2Protect [5], authors manipulated facial attributes to perform adversarial attacks. In Adv-Makeup [8], the authors craft transferable black-box adversarial faces via task-driven makeup generation with meta-learning. In AMT-GAN [4], the authors improve black-box transferability while preserving visual realism through regularization and joint training. In Adv-Attribute [23], the authors enhance transferability by generating attribute-specific perturbations that target high-level face semantics. In Clip2Protect [5], the authors found adversarial latent codes in the low dimensional manifold of a pretrained generative model and fine-tunes the generative model to achieve an accurate reconstruction of the given image from its latent code. However, as it is hard to balance the gradient from the original face image and the target FR model, the GAN-based methods remain obvious artifacts in adversarial face images.

In AdvDiffusion [7], the authors generate transferable imperceptible perturbations in diffusion latent space via conditional diffusion. In ADA [24], the authors improve diffusion-based attacks by dynamically balancing perturbation injection and denoising in reverse diffusion to boost robustness and image quality. However, SD-based attacks often blend original and target identity features, causing noticeable distortions. A key limitation is the unreliable transferability assumption: substitute face recognition models, due to architectural and data differences—fail to accurately mimic the target FR model, leading to high success on substitutes but poor performance on actual targets, significantly reducing both attack effectiveness and image quality.

2) *Query-Based Adversarial Attacks*: Query-based adversarial attacks have gained significant attention due to their practicality in real-world scenarios, where the attackers often face black-box settings with limited access to model parameters and gradients. These methods rely solely on queries to the target FR model, typically using top-1 predicted labels or score-based replies to craft adversarial examples.

In Evolution [25], the authors improve query efficiency by modeling local search-direction geometry to reduce perturbation size. In IDBA [26], the authors enhance imperceptibility and efficiency by optimizing perceptually guided distributions that target decision-relevant components. In CGBA [27], the authors accelerate boundary-based black-box attacks via curvature-aware boundary search. In Sparse-RS [28], the authors enable query-efficient black-box attacks under sparse perturbation constraints (e.g., patches/frames). In BruSLeAttack [29], the authors use Bayesian optimization for sparse-sample attacks to balance query cost and attack strength. In DevoPatch [30], the authors improve query-efficient targeted/untargeted patch attacks via evolutionary optimization. In GSBAK [31], the authors craft query-efficient score-based attacks via gradient-initialized boundary points

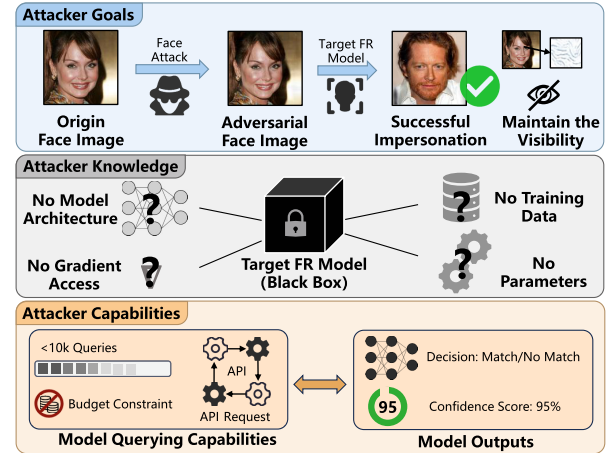


Fig. 2. Threat model.

and boundary-following updates, extending to targeted or untargeted and multi-label settings.

Despite their advantages, query-based attacks face several challenges. Firstly, the number of queries required to craft high-quality adversarial examples remains a bottleneck as excessive queries can lead to detection by defense systems, potentially limiting their practicality in security-sensitive applications. Secondly, black-box settings impose significant constraints on available information, making it difficult to achieve optimal perturbations with minimal distortion. Addressing these challenges is critical for improving real-world applicability of query-based adversarial attacks.

III. THREAT MODEL

We describe the threat model from the perspectives of an attacker's goal, background knowledge, and capabilities, as illustrated in Figure 2.

A. Attacker's Goals

The goal of the attacker is to add adversarial perturbations to face images which can cause the target FR model (even with adversarial defense strategies) to recognize the adversarial face images as the target identity, while not significantly affecting the visibility of the original input.

B. Attacker's Background Knowledge

The attacker has no knowledge on the target FR model, including its architecture, parameters, configurations, training data. This means the attacker cannot obtain the behavior of the target FR model or its gradient.

C. Attacker's Capabilities

The attacker can collect face images of the enrolled target identities, which is feasible in reality, for example, through social platforms and social engineering techniques. The attacker can repeatedly query the target FR model to obtain the recognition result. In practice, the adversary has constraints on computational resources. Online services also

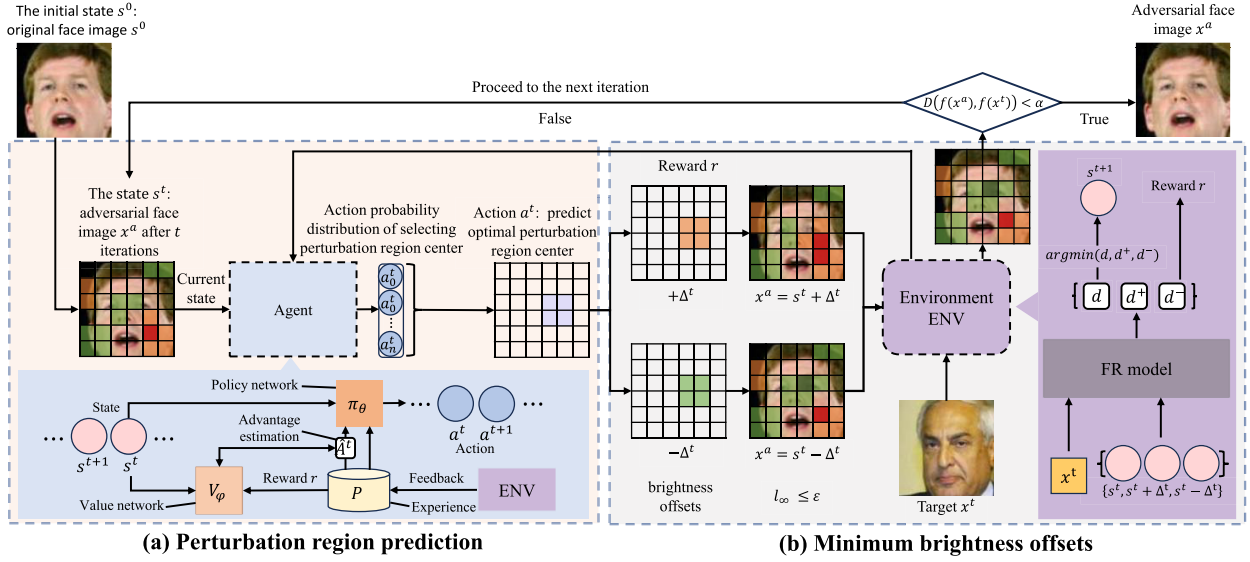


Fig. 3. The architecture of PPOM-ATTACK. The (a) illustrates the process of PPO agent-based perturbation region prediction and the (b) exhibits process of minimum brightness offsets optimization. The PPO agent predicts the region to be perturbed based on the adversarial face image $x^a \sim s^t$ from the t^{th} iteration (see in Section V-B). Then, PPOM-ATTACK applies a minimal brightness offset in the predicted perturbation region to minimize the distance between the feature embedding of the adversarial face image x^a and the target face image x^t (see in Section V-C).

prevent excessive model queries. We consider a budget for the number of queries 10k to carry out the attack. The attacker can only access to the recognition result (decision result and confidence scores) of a target FR model for each test input.

IV. PROBLEM FORMULATION

PPOM-ATTACK aims to make the target FR model recognize the original identity as the other one while minimizing the perturbations added to the original face image.

We formulate the adversarial face image generation problem as an optimization problem, as shown in Eq. (1), where x^a is the adversarial face image, x^t is the targeted face image and x^o is the original face image. The objective is to minimize the distance $D(f(x^a), f(x^t))$ between the feature embedding of the adversarial face image $f(x^a)$ and the feature embedding of the target face image $f(x^t)$ while restricting the l_∞ norm between the x^a and x^o to a small constant ϵ . The distance $D(f(x^a), f(x^t))$ is the feedback of the target FR model to measure the similarity of x^a and x^t . The distance decreases as the adversarial face image is more likely to be recognized as the target individual by the FR model.

$$\begin{aligned} \text{obj} : \min_{x^a} D(f(x^a), f(x^t)), \\ \text{s.t. } l_\infty = \|x^a - x^o\|_\infty \leq \epsilon \end{aligned} \quad (1)$$

We formulate the adversarial face image perturbation according to Eq. (2) and Eq. (3), where δ is the perturbation added to the original face image x^o , and Δ^t represents the perturbation in the t^{th} iteration.

$$x^a = x^o + \delta \quad (2)$$

$$\delta = \pm \Delta^t \pm \Delta^{t-1} \pm \dots \pm \Delta^0 \quad (3)$$

PPOM-ATTACK predicts and optimizes the δ to reach the objective function and satisfy the constraint.

V. METHODOLOGY

A. The Framework of PPOM-ATTACK

The PPOM-ATTACK is a substitute model-free perturbation prediction and optimization method for black-box adversarial attacks against FR models. Unlike previous approaches that rely on substitute models, PPOM-ATTACK directly obtains feedback from the target model, thus avoiding any discrepancies with the attack objective. The architecture of our approach is presented in Figure 3. It includes two parts: PPO agent-based perturbation region prediction and Minimum brightness offsets optimization. This architecture forms a hierarchical perturbation strategy, where the agent operates on region level actions while their effects propagate to pixel level modifications.

We define a perturbation region as a contiguous square patch centered at a pixel location in the face image. For an image of width w , the side length of the patch is set to $\sqrt{\rho w^2}$ so that it covers approximately a fraction ρ of the pixels; if the center lies near the boundary, the patch is cropped to remain within the image domain.

- In the PPO agent-based perturbation region prediction method, a PPO-based agent is designed to predict perturbation regions in the face image and self-adaptively disturb these regions. The agent observes the current state of the disturbed face image and, based on this state, predicts the probability of each perturbation region within the image. It then samples perturbation regions according to the predicted probability distribution and disturbs the sampled regions.
- To maintain high-quality adversarial face images, we introduce a minimum brightness offsets method specifically designed to generate perturbations (i.e., ± 1 for the entire region) and add them to the sampled perturbation region. We then update the current disturbed face image

Algorithm 1 PPOM-ATTACK

```

1: Input: original image  $x^o$ , target image  $x^t$ , FR model  $f$ ,
   initial policy network  $\pi_\theta$ , value network  $V_\phi$ , maximum
   number of iterations  $N$ , update frequency  $T$ , region size
    $\rho$ , perturbation budget  $\epsilon$ , stopping threshold  $\alpha$ .
2: Output: adversarial facial image  $x^a$ , query count  $q$ .
3: Compute normalized embeddings  $\mathbf{e}^0 \leftarrow \text{Norm}(f(x^o))$ ,
    $\mathbf{e}^{\text{tar}} \leftarrow \text{Norm}(f(x^t))$ .
4:  $d \leftarrow \|\mathbf{e}^0 - \mathbf{e}^{\text{tar}}\|_2$ ,  $s^0 \leftarrow x^o$ ,  $t \leftarrow 0$ ,  $q \leftarrow 1$ .
5:  $x_{\min} \leftarrow \text{clip}(x^o - \epsilon, 0, 255)$ ,  $x_{\max} \leftarrow \text{clip}(x^o + \epsilon, 0, 255)$ .
6: Initialize experience pool  $P \leftarrow \emptyset$ .
7: while  $d \geq \alpha$  and  $t < N$  do
8:   // PPO agent-based Perturbation Region Prediction
   Phase
9:   Current state  $s^t$  corresponds to the current adversarial
   image.
10:  Predict perturbation region probabilities using  $\pi_\theta(s^t)$ .
11:  Sample a discrete region center  $a^t \sim \pi_\theta(\cdot | s^t)$ , where
    $a^t \in \{1, \dots, w^2\}$ .
12:  Map  $a^t$  to 2D coordinates  $(x, y)$  and define perturbation
   region  $R(a^t)$  of size  $\sqrt{\rho * w^2} \times \sqrt{\rho * w^2}$ .
13:  // Minimum Brightness Offsets Optimization Phase
14:  Form candidate state  $s^{t,+}$  by adding  $\Delta^t$  on  $R(a^t)$  of  $s^t$ 
   and clipping to  $[x_{\min}, x_{\max}]$ .
15:  Form candidate state  $s^{t,-}$  by subtracting  $\Delta^t$  on  $R(a^t)$  of
    $s^t$  and clipping to  $[x_{\min}, x_{\max}]$ .
16:  Compute  $\mathbf{e}^{t,+} \leftarrow \text{Norm}(f(s^{t,+}))$ ,  $\mathbf{e}^{t,-} \leftarrow \text{Norm}(f(s^{t,-}))$ .
17:   $d^+ \leftarrow \|\mathbf{e}^{t,+} - \mathbf{e}^{\text{tar}}\|_2$ ,  $d^- \leftarrow \|\mathbf{e}^{t,-} - \mathbf{e}^{\text{tar}}\|_2$ ,  $q \leftarrow q + 2$ .
18:  if  $d^- < d^+$  then
19:     $s' \leftarrow s^{t,-}$ ,  $d' \leftarrow d^-$ .
20:  else
21:     $s' \leftarrow s^{t,+}$ ,  $d' \leftarrow d^+$ .
22:  end if
23:  if  $d' \geq d^t$  then
24:     $s^{t+1} \leftarrow s^t$ ,  $d^{t+1} \leftarrow d^t$  // Reject non-improving
    perturbation
25:  else
26:     $s^{t+1} \leftarrow s'$ ,  $d^{t+1} \leftarrow d'$ .
27:  end if
28:   $r^t \leftarrow \beta(\alpha - d^{t+1}) + (d^t - d^{t+1})$ .
29:  Compute  $\log \pi_\theta(a^t | s^t)$  and  $v^t \leftarrow V_\phi(s^t)$ .
30:  Store  $(s^t, a^t, r^t, s^{t+1})$  into pool  $P$ .
31:  // On-Policy Training Phase
32:  if  $t \bmod T = 0$  and  $P \neq \emptyset$  then
33:    Use experiences in  $P$  to update  $\pi_\theta$  and  $V_\phi$ .
34:    Clear experience pool  $P$ .
35:  end if
36:   $t \leftarrow t + 1$ ,  $d^t \leftarrow d^{t+1}$ .
37: end while
38:  $x^a \leftarrow s^t$  // Final adversarial facial image
39: return  $x^a, q$ .

```

with the sampled perturbation that performs best in the current round, that minimize the feature embedding difference between the adversarial and targeted face images.

The process of PPOM-ATTACK is presented in Algorithm 1. The process begins with the Initialization

TABLE I

SUMMARY OF MAIN NOTATION USED IN PPOM-ATTACK

Symbol	Description
x^o	Original (benign) face image.
x^t	Target face image.
x^a	Adversarial face image.
s^t	State at iteration t (current adversarial image, with $s^0 = x^o$).
a^t	Action at iteration t (center of the selected perturbation region).
w	Image width/height (assuming square face images).
ρ	Maximum fraction of pixels allowed to be modified in the face image.
Δ^t	Brightness offset matrix applied to the selected region at iteration t .
$f(\cdot)$	Target face recognition model that outputs feature embeddings.
$D(\cdot, \cdot)$	Distance function between two feature embeddings.
d^t	Embedding distance $D(f(s^t), f(x^t))$ at iteration t .
r^t	Immediate reward obtained at iteration t .
R^t	Discounted return starting from iteration t .
π_θ	Policy network (agent) parameterized by θ .
V_ϕ	Value network parameterized by ϕ .
A^t	Advantage estimate at iteration t .
γ	Discount factor for future rewards.
λ	GAE parameter balancing bias and variance.
η	PPO clipping threshold in the surrogate objective.
α	Distance threshold deciding whether s^t and s^t are recognized as the same identity.
β	Hyper-parameter balancing rewards and penalties in the reward function.
T	Number of environment steps in one PPO update (trajectory length / update interval).
P	Experience pool storing (s^t, a^t, r^t, s^{t+1}) tuples.

Phase, where the original face image x^o is designated as the initial state s^0 , and the target face image x^t is defined. At this stage, the parameters of policy and value networks are initialized, and the initial feature embedding distance between the original and target images is calculated. In the PPO agent-based Perturbation Region Prediction Phase, the policy network observes the current state s^t of the adversarial face image x^a and take action a^t to predict a probability distribution over perturbation region centers in the current state s^t of adversarial face image x^a , with higher probabilities assigned to areas likely to reduce the feature embedding distance between the current state s^t of adversarial face image x^a and the target face image x^t when perturbed. Moving into the Minimum Brightness Offsets Optimization Phase, perturbation region centers are sampled from the predicted distribution, and for each sampled region, three candidate perturbations: no offset, adding uniform positive or negative brightness offsets are applied to the region while ensuring all pixel values respect predefined constraints. The feature embedding distance to the target image x^t is then computed for each candidate, and the perturbation that minimizes this distance is selected to update the adversarial face image x^a as the current state s^t , followed by reward r calculation based on the distance reduction. During the On-Policy Training Phase, the resulting experience tuple (s^t, a^t, r, s^{t+1}) is stored in an experience pool P , and every T iterations, the policy and value networks are updated using the accumulated experiences to improve the policy network's ability to predict effective perturbation region centers. Finally, the algorithm cycles through the PPO agent-based Perturbation Region Prediction Phase, the Minimum Brightness Offsets Optimization Phase, and the On-Policy Training Phase until the current state s^t of adversarial face image x^a successfully deceives the target FR model or the maximum number of iterations is reached, upon which the algorithm outputs the final adversarial face image x^a .

To make the subsequent reinforcement learning formulation, loss functions, and reward definition easier to follow, we summarize the main notation used in PPOM-ATTACK in Table I.

B. PPO Agent-Based Perturbation Region Prediction

Unlike the substitute model-based generative methods, in this work, one major challenge is the lack of gradient

information in feedback from the target FR model, leading to inefficient adversarial face image generation and potentially random perturbations. To address this issue, we propose a PPO-based agent (hereafter referred to as “the agent”) that predicts perturbation regions and adaptively disturbs these regions. We model the process of adding perturbations as a Markov Decision Process (MDP), in which we design the agent’s state, action, and reward, and subsequently train its policy and value networks using these elements.

In the MDP, the agent perceives the current state of the perturbed adversarial face image and, based on this perception, predicts the probability for each perturbation regions within the image. It then samples regions for perturbation according to the predicted probability distribution and applies noise (see in Section V-C) to the sampled regions. Through this iterative process, the agent learns to optimize its actions based on feedback from the target FR model, even in the absence of explicit gradient information, to generate adversarial face images.

1) *State Encoding*: The state s perceived by the agent is the state of the adversarial image x^a , with the original image x^o serving as the initial state s^0 . As the process evolves, the state s^t is updated by applying perturbations according to Eq. (4) to the selected perturbation regions in the t^{th} iteration, reflecting the current state of the adversarial image x^a .

$$s^t = s^{t-1} \pm \Delta^{t-1} \quad (4)$$

2) *Action Encoding*: We define actions a^t as the selection of perturbation region centers within the image in the t^{th} iteration. Considering that any pixel coordinates (x, y) in the image $I(x, y)$ could serve as the center of a perturbation region, we denote $a^t \in (x, y)$, where (x, y) represents the x-coordinates and y-coordinates of these center points.

3) *Policy and Value Networks*: In the agent, we employ a policy network to perceive and predict the perturbation regions, and a value model to assist in the training of the policy network. The Convolution Neural Network (CNN) used in our approach to train the policy and the value model consists of two convolution layers and three fully connected layers. The output of the policy model is the probability of perturbation region centers in the image. The output of the value model is a value used to estimate the value of the current state s^t .

4) *RL Formulation and Choice of PPO*: The facial recognition adversarial attack process in PPOM-ATTACK can be modeled as a Markov decision process (MDP) with a large action space. At each step, the agent selects a perturbation region and updates the adversarial image, which immediately changes the subsequent state distribution. This leads to a highly non-stationary state distribution under on-policy learning and makes many standard value-based or off-policy algorithms, such as DQN, TD3 methods, less suitable. Value-based methods and their variants are ineffective in this setting for two main reasons. First, the large action space makes reliable estimation of $Q(s, a)$ and the $\arg \max_a Q(s, a)$ selection prohibitively expensive: evaluating only a subset of actions introduces bias, while evaluating all actions at every step is infeasible under a strict query budget. Second, most practical value-based algorithms are implemented in an off-policy

manner with replay buffers and temporal-difference bootstrapping. In our attack formulation, however, the state distribution depends strongly on the current policy and changes after every perturbation, so replayed transitions quickly become mismatched with the current policy. This off-policy mismatch degrades Q function estimates and leads to unstable learning under a limited query budget.

We therefore adopt proximal policy optimization (PPO) as the backbone of PPOM-ATTACK. PPO is an on-policy policy-gradient method that directly optimizes a stochastic policy over the discrete set of perturbation regions and is widely used in problems with large action spaces. Recent PPO variants (e.g., GRPO, PPG, KL-penalized PPO) introduce more elaborate training schemes to obtain moderate empirical gains, but typically require additional passes, auxiliary losses, or tighter hyper-parameter tuning. Under our query-limited black-box attack setting, such extra complexity would increase computational and implementation overhead without changing the interaction budget with the target model. To balance stability and efficiency, PPOM-ATTACK employs the standard PPO framework with mini-batch optimization and multiple epochs over each batch of on-policy data, and parameterizes the policy as a probability distribution over all candidate perturbation regions in the large action space.

5) *Training*: We propose a PPO agent on-policy training method to train the policy network. First, we initialize the state s^0 as x^o and obtain the initial distance $D(f(x^o), f(x^t))$ between the feature embedding $f(x^o)$ of the initial state s^0 and the feature embedding $f(x^t)$ of the target face image x^t . Then we take an action a to select the perturbation region predicted by the policy network π_θ . In our method, we do not need to pre-train a model with a training data set. We only train the policy and value models by self-learning. For each update of the policy network π_θ and the value network V_ϕ , we use data from the experience pool. And we clear the experience pool after each update.

During training, there existed several practical challenges of stability and convergence to be solved. For the challenge of the large action space and spatial redundancy, our approach uses clipped policy updates and gradient clipping to reduce the variance of the policy gradient. For the challenge of the non-stationary state distribution, our approach uses on-policy training with re-sampling in mini-batch updates to avoids distributional drift caused by iterative image perturbation. For the challenge of the local optima, our approach uses entropy regularization to encourages sufficient exploration and prevents premature collapse in the large action space.

The loss function for the policy network is designed to optimize the policy network that it maximizes the expected cumulative reward of the adversarial perturbation. Specifically, the loss function can be expressed as Eq. (5), similar to [32].

$$\begin{aligned} \underset{\theta}{\text{maximize}} L^{CLIP}(\theta) = & \frac{1}{T} \sum_{t=0}^T \left[\min \left(\frac{\pi_\theta(s^t, a)}{\pi_{\theta_{old}}(s^t, a)} \hat{A}^t, \right. \right. \\ & \left. \left. \text{clip} \left(\frac{\pi_\theta(s^t, a)}{\pi_{\theta_{old}}(s^t, a)}, 1 - \eta, 1 + \eta \right) \hat{A}^t \right) \right] \end{aligned} \quad (5)$$

where $\text{clip}()$ is the clip function to avoid too large difference between the new and old policies, with the clip range $1 \pm \eta$ ($\eta = 0.2$), and $\hat{A}^t$ represents an estimator of the advantage function at iteration t , which takes the following Eq. (6).

$$\hat{A}^t = \sum_{l=0}^T (\gamma \lambda)^l (r^t + \gamma V_\phi(s^{t+l+1}) - V_\phi(s^{t+l})) \quad (6)$$

where r^t represents the reward value for taking action a in the current state. $V_\phi()$ represents the expected future rewards according to the current state s^t , which is also known as the value v of the current state s^t . γ ($\gamma = 0.99$) is the discount factor used to balance the weight of future rewards and current rewards. λ ($\lambda = 0.95$) is a hyper-parameter used to balance variance and bias. For the value network V_ϕ , we use Eq. (7) as the loss function to update the value network, where R_t is the final reward and $V_\phi(s^t)$ is the predict value of s^t calculated by the value network.

$$\underset{\phi}{\text{minimize}} L^V(\phi) = \frac{1}{T} \sum_{t=0}^T (R^t - V_\phi(s^t))^2 \quad (7)$$

In addition to the above losses, we apply entropy regularization and gradient clipping (0.5) during optimization to prevent premature policy collapse and stabilize training under the non-stationary dynamics caused by iterative image perturbation.

C. Minimum Brightness Offsets Optimization

Next, after selecting the perturbation region by the agent, we apply a single-step brightness offset perturbation to the image to obtain the next state s^{t+1} of the adversarial face image x^a . We then calculate the feature embedding distance $D(f(s^{t+1}), f(x^t))$ between the next state s^{t+1} and the target face image x^t , and compute the reward value r (see in Section V-C). We store (s^t, a, r, s^{t+1}) in the experience pool for training of π_θ and V_ϕ . For each T ($T = 10$) steps, we update policy network π_θ and value network V_ϕ with the data from the experience pool. Once done, we proceed to the next iteration until either the maximum number of iterations is reached or the adversarial face image successfully attacks the target FR model. Finally, we stop the iteration and obtain the adversarial face image x^a as the output.

In the optimization process, we propose a minimum brightness offsets optimization method which first uses an all ones matrix $+\Delta^t$ and an all negative ones matrix $-\Delta^t$ as brightness offsets for disturbing the sampled perturbation region. For the current state s^t of adversarial face image x^a , we add perturbation Δ^t to the perturbation region selected by policy network π_θ of the agent. Then the disturbed image will become to the next state s^{t+1} of adversarial face image x^a . The method finally calculates the reward r and sends back to the PPO-based agent to update the policy and value networks.

1) *Brightness Offsets*: We propose a all ones matrix and a all negative ones matrix as the brightness offsets. The size of the matrix is $\sqrt{\rho * w^2} \times \sqrt{\rho * w^2}$, as the same as the size of the perturbation region, where $\rho \in (0, 1)$ represents the maximum percentage of the modified pixels in the face image and w is the size of the face image.

Because all elements in the offset matrix share the same value within the selected square region, each update corresponds to a uniform brightness change on a contiguous region of size $\sqrt{\rho * w^2} \times \sqrt{\rho * w^2}$. This region-wise parameterization restricts the perturbation to spatially coherent areas instead of isolated pixels. In practice, such local brightness adjustments appear as mild illumination variations in the face image, while the rest of the image remains unchanged, better approximating the variation patterns observed in natural images.

Once the perturbation region is selected by the policy network π_θ , we need to truncate it to the valid range $[0, w - 1]$ to avoid exceeding image boundary. Then, we add the all ones matrix to the pixel values $x^a \sim s^{t+1} = s^t + \Delta^t$ and the all negative ones matrix to the pixel values $x^a \sim s^{t+1} = s^t - \Delta^t$ in the perturbation region respectively as the candidate perturbations. Meanwhile, we also need to ensure that the disturbed pixel value satisfies the maximum pixel value boundary and a hard constraint $l_\infty \leq \epsilon$, as shown in Eq. (8).

$$x^a \in [\max(0, x^o - \epsilon), \min(x^o + \epsilon, 255)] \quad (8)$$

2) *Optimal Perturbation and Reward Calculation*: Next, for each selected region, we consider three candidate updates: no change, adding $+\Delta^t$, and adding $-\Delta^t$ under the ℓ_∞ constraint. For each candidate, we compute the Euclidean distance of embedding to the target image, i.e., $D(f(s^t), f(x^t))$, $D(f(s^t + \Delta^t), f(x^t))$, and $D(f(s^t - \Delta^t), f(x^t))$. We then select the candidate that yields the smallest embedding distance as the optimal perturbation, as summarized in Eq. (9).

$$\begin{aligned} d^{t+1} = \min & (D(f(s^t + \Delta^t), f(x^t)), \\ & D(f(s^t - \Delta^t), f(x^t)), \\ & D(f(s^t), f(x^t))) \end{aligned} \quad (9)$$

Because each candidate modifies the entire region in a uniform manner, the resulting updates remain spatially smooth, and the optimization procedure avoids accumulating uneven or visually disruptive changes within the face. Regions whose updates consistently lead to larger distances naturally receive lower returns and are less likely to be selected by the agent in subsequent iterations.

Finally, we update the image with the optimal perturbation, and calculate the reward r designed by reward shaping for the current disturbed image.

$$\text{reward}(d^t, d^{t+1}, \alpha) = \beta \cdot (\alpha - d^{t+1}) + (d^t - d^{t+1}) \quad (10)$$

VI. EXPERIMENT

A. Experimental Setting

1) *Setup*: We implemented all of the methods in the same environment and configuration. The CPU is Intel i9-13900k, with 64.0GB of RAM, and the GPU is RTX 4090 24.0GB. All of the methods are implemented in PyTorch framework.

2) *Parameter Setting*: The hyper-parameters ϵ and β are set to be 15, 0.001 respectively. We train the policy network and the value network by an Adam optimizer with the learning rate of 0.0003, and set exponential decay rates as (0.9, 0.999).

3) *Datasets*: We conduct experiments on three face datasets: Labeled Faces in the Wild (LFW) [33], CelebA-HQ [34], and VGGFace2 [35]. LFW is a standard face recognition benchmark that contains 13,233 images of 5,749 subjects collected from the web. CelebA-HQ is a high-quality version of the CelebA dataset, consisting of 30,000 frontal and near-frontal facial images with higher resolution and cleaner artifacts. VGGFace2 is a large-scale face dataset with a wide range of pose, illumination, expression, age, and occlusion variations, providing a more challenging and diverse evaluation setting than LFW and CelebA-HQ.

Since our approach does not train any substitute model or auxiliary pre-trained network, no additional training data are required for PPOM-ATTACK. The comparison methods use publicly available pre-trained face recognition models as in their original papers. For evaluation, we randomly sample 1500 source–target identity pairs from LFW, CelebA-HQ, and VGGFace2 for our method and other comparison methods.

In our implementation, the policy and value networks of PPOM-ATTACK are updated online during the attack process using only the trajectories collected from queries to the target FR model. All selected images from LFW, CelebA-HQ, and VGGFace2 are therefore used purely as test data, and no dataset is used for offline training of the attack.

4) *Baselines*: We implement several famous black-box adversarial attack methods for comparison, including PGD [1], TI-RDIM [2], TIP-IM [3], Clip2Protect [5], Sibling-Attack [17], ATT [22], AdvDiffusion [7], DPA [21], Evolution [25], CGBA [27], DevoPatch [30], Sparse-RS [28], BruSLeAttack [29] and GSBak [31].

5) *Target Models*: Similar to the evaluation in prior works [7], we conduct experiments to attack 7 famous FR models as the target black-box models, which include IR50 [36], Mobileface [37], Curricularface [38], Insightface [39], Shuffleface [40], Arcface [41] and Centerloss [42]. Additionally, we evaluate experiments to attack two FR models with adversarial defense strategies: PGD-AT [1] and TRADES [43], as well as the widely-used commercial FR model, Face++.

6) *Substitute Models*: We conduct the substitute model for the baseline methods against the target FR models. For each target FR model, the other FR models will serve as the ensemble substitute model. In the experiments attacking Face++ and robust FR models, the seven FR models will serve as the ensemble substitute model. Notably, Sibling-Attack uses its own trained multi-task IR152 model as the substitute model. DPA uses Diverse Parameters Optimization (DPO) method trained the parameter diversity substitute models based on MobileFace. ATT uses pre-training Transface [44] as the substitute model.

7) *Evaluation Metrics*: We use Attack Success Rate (ASR) to evaluate the attack ability of different methods, according to Eq. (11):

$$ASR = \frac{\text{No. of Comparisons} \leq \alpha}{\text{Total No. of Comparisons}} \quad (11)$$

We calculate the ASR at False Acceptance Rate (FAR)@0.01 for black-box testing [7]. The value of α will be set as the threshold at FAR@0.01 for each victim FR model as most

works do, i.e., IR50 ($\alpha = 1.2092$), Mobileface ($\alpha = 1.1727$), Curricularface ($\alpha = 1.2862$), Insightface ($\alpha = 1.2701$), Shuffleface ($\alpha = 0.9253$), Arcface ($\alpha = 1.1611$) and Centerloss ($\alpha = 0.6691$). We also leverage Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) to evaluate the image quality as most related works do. Additionally, we use the l_2 norm as a measure of the perturbation magnitude. To further assess the perceptual fidelity of adversarial images, we incorporate two learned perceptual metrics: Learned Perceptual Image Patch Similarity (LPIPS) and Deep Image Structure and Texture Similarity (DISTS). LPIPS measures the distance between deep feature representations extracted from pretrained networks and correlates strongly with human perceptual judgments, that may not be reflected in PSNR and SSIM. DISTS jointly models high-level structural information and low-level texture statistics, providing a more comprehensive perceptual similarity assessment that is robust to illumination or contrast variations.

B. Experimental Evaluation

1) *Experimental Design*: In this section, we show experimental results cross different data sets as LFW dataset and CelebA-HQ dataset. We conducted state-of-the-art (SOTA) adversarial face attack methods including PGD, TI-RDIM, TIP-IM, Clip2Protect (CVPR 2023), Sibling-Attack (CVPR 2023), ATT (NIPS 2024), AdvDiffusion (AAAI 2024), DPA (CVPR 2025) and Our method PPOM-ATTACK on widely used FR models including IR50, Mobileface, Curricularface, Insightface, Shuffleface, Arcface and Centerloss and commercial FR model Face++ respectively. In our PPOM-ATTACK, we set the $\rho = 0.02$, which controls the size of the perturbation region. For the SOTA attack methods, we used default settings. We evaluate the ASR, PSNR, SSIM, l_2 norm, LPIPS and DISTS of adversarial face images generated by each attack method cross 8 FR models (including Face++). What's more, we compare the ASR of our PPOM-Attack method with performance-stronger attack methods such as Clip2Protect, Sibling-Attack and AdvDiffusion with adversarial trained defence strategy. To further validate the effectiveness of PPOM-ATTACK, we extended our experiments to query-based attack methods, where we compared with existing decision-based and score-based approaches, including Evolution (CVPR 2019), CGBA (ICCV 2023) and DevoPatch (TIFS 2023) for decision-based, and Sparse-RS (AAAI 2022), BruSLeAttack (ICLR 2024) and GSBak (ICLR 2025) for score-based attacks. We paid more attention to the number of queries. For a fair evaluation, the maximum number of allowed queries for all query-based attack methods was set to 10,000 or close to 10,000 (such as CGBA).

2) *Evaluation of Substitute Model-Based Adversarial Face Attack Methods*: We compare the ASR, PSNR, SSIM l_2 norm, LPIPS and DISTS for each methods cross two different data sets, as shown in Table II.

- In the experimental results on the LFW dataset, PPOM-ATTACK achieves a higher ASR compared to SOTA methods. Specifically, PPOM-ATTACK outperforms the best-performing method for each target FR model by

TABLE II
PERFORMANCE COMPARISON WITH SOTA SUBSTITUTE MODEL-BASED METHODS ON LFW AND CELEBA-HQ DATASETS

Metric	Attack method / FR model	IR50 LFW / CelebA-HQ	MobileFace LFW / CelebA-HQ	CurricularFace LFW / CelebA-HQ	InsightFace LFW / CelebA-HQ	ShuffleFace LFW / CelebA-HQ	Arcface LFW / CelebA-HQ	Centerloss LFW / CelebA-HQ
ASR(%)	PGD	64.0 / 40.8	65.2 / 43.2	96.2 / 82.6	97.2 / 78.6	11.2 / 2.6	77.6 / 45.6	12.0 / 7.6
	TI-RDIM	42.6 / 17.4	51.6 / 38.0	62.6 / 32.4	77.8 / 48.0	3.4 / 0.2	32.0 / 17.2	9.6 / 6.2
	TIP-IM (IJCAI 2021)	36.0 / 14.4	48.8 / 22.8	67.6 / 24.6	76.6 / 29.4	4.8 / 1.4	55.0 / 12.6	8.2 / 6.8
	Clip2Protect (CVPR 2023)	71.4 / 17.8	74.4 / 44.0	86.6 / 39.4	91.4 / 48.0	8.0 / 2.4	76.6 / 30.6	18.2 / 5.4
	Sibling-Attack (CVPR 2023)	99.0 / 49.0	99.0 / 70.6	99.0 / 70.6	98.6 / 77.4	9.0 / 1.4	94.8 / 57.2	11.8 / 6.4
	ATT (NIPS 2024)	48.0 / 15.6	38.6 / 25.4	97.6 / 69.4	99.4 / 78.6	2.8 / 0.2	39.4 / 17.2	5.4 / 3.0
	AdvDiffusion (AAAI 2024)	62.0 / 70.2	64.4 / 73.0	78.6 / 90.6	81.4 / 92.8	7.6 / 12.8	65.2 / 74.2	21.8 / 22.2
	DPA (CVPR 2025)	41.6 / 26.0	100.0 / 87.2	67.0 / 44.2	79.2 / 50.2	38.6 / 13.6	95.6 / 76.8	24.8 / 8.4
	PPOM-ATTACK (Ours)	100.0 / 87.0	100.0 / 89.4	94.8 / 96.2	100.0 / 91.4	99.6 / 64.8	94.6 / 96.8	67.0 / 70.4
SSIM(↑)	PGD	0.8954 / 0.8975	0.9031 / 0.9025	0.8984 / 0.8999	0.8986 / 0.8999	0.8843 / 0.8980	0.8857 / 0.9001	0.8857 / 0.8837
	TI-RDIM	0.6692 / 0.7089	0.6660 / 0.7082	0.6702 / 0.7103	0.6694 / 0.7100	0.6679 / 0.7083	0.6693 / 0.7109	0.6669 / 0.7075
	TIP-IM (IJCAI 2021)	0.8856 / 0.8847	0.8622 / 0.8865	0.8614 / 0.8848	0.8613 / 0.8848	0.8629 / 0.8851	0.8646 / 0.8869	0.8613 / 0.8895
	Clip2Protect (CVPR 2023)	0.2584 / 0.5977	0.2592 / 0.6011	0.2587 / 0.5995	0.2578 / 0.6021	0.2578 / 0.5995	0.2587 / 0.5963	0.2600 / 0.5991
	Sibling-Attack (CVPR 2023)	0.6565 / 0.6908	0.6565 / 0.6908	0.6565 / 0.6908	0.6565 / 0.6908	0.6565 / 0.6908	0.6565 / 0.6908	0.6565 / 0.6908
	ATT (NIPS 2024)	0.6817 / 0.7455	0.6817 / 0.7455	0.6817 / 0.7455	0.6817 / 0.7455	0.6817 / 0.7455	0.6817 / 0.7455	0.6817 / 0.7455
	AdvDiffusion (AAAI 2024)	0.4517 / 0.4801	0.4565 / 0.4810	0.4529 / 0.4797	0.4541 / 0.4793	0.4521 / 0.4818	0.4549 / 0.4793	0.4588 / 0.4779
	DPA (CVPR 2025)	0.8172 / 0.8242	0.8172 / 0.8242	0.8172 / 0.8242	0.8172 / 0.8242	0.8172 / 0.8242	0.8172 / 0.8242	0.8172 / 0.8242
	PPOM-ATTACK (Ours)	0.9616 / 0.9717	0.9643 / 0.9726	0.9758 / 0.9764	0.9743 / 0.9761	0.9640 / 0.9381	0.9681 / 0.9684	0.9368 / 0.9599
PSNR(↑)	PGD	32.4600 / 30.6808	32.8801 / 30.8762	32.6069 / 30.7802	32.6383 / 30.7978	32.1077 / 30.7267	32.5704 / 30.7624	32.0056 / 30.2628
	TI-RDIM	24.3505 / 24.5532	24.3470 / 24.5539	24.3432 / 24.5588	24.3383 / 24.5540	24.3434 / 24.5549	24.3440 / 24.5505	24.3400 / 24.5565
	TIP-IM (IJCAI 2021)	31.1682 / 30.1981	31.3233 / 30.2644	31.2546 / 30.2017	31.2570 / 30.2025	31.1130 / 30.1789	31.3994 / 30.2718	31.2503 / 30.3048
	Clip2Protect (CVPR 2023)	12.5459 / 18.5468	12.5798 / 18.6200	12.5478 / 18.5725	12.5473 / 18.6165	12.5405 / 18.5227	12.5669 / 18.5408	12.5922 / 18.5313
	Sibling-Attack (CVPR 2023)	25.4783 / 25.5051	25.4783 / 25.5051	25.4783 / 25.5051	25.4783 / 25.5051	25.4783 / 25.5051	25.4783 / 25.5051	25.4783 / 25.5051
	ATT (NIPS 2024)	26.4728 / 27.2097	26.4728 / 27.2097	26.4728 / 27.2097	26.4728 / 27.2097	26.4728 / 27.2097	26.4728 / 27.2097	26.4728 / 27.2097
	AdvDiffusion (AAAI 2024)	14.2092 / 15.1482	14.4135 / 15.1488	14.3668 / 15.1205	14.4062 / 15.1321	14.3460 / 15.1846	14.3980 / 15.0933	14.5040 / 15.1127
	DPA (CVPR 2025)	29.7577 / 29.8215	29.7577 / 29.8215	29.7577 / 29.8215	29.7577 / 29.8215	29.7577 / 29.8215	29.7577 / 29.8215	29.7577 / 29.8215
	PPOM-ATTACK (Ours)	33.2540 / 34.2912	33.8222 / 35.2824	35.9093 / 35.4276	35.7524 / 35.4713	33.9150 / 32.7833	34.5162 / 33.6735	31.3367 / 31.5442
$l_2(\downarrow)$	PGD	1193.9301 / 1673.9674	1137.5677 / 1637.7285	1173.8951 / 1655.3914	1169.6854 / 1651.9736	1245.8025 / 1665.4353	1178.7849 / 1658.8689	1257.7890 / 1753.1618
	TI-RDIM	2998.1886 / 3347.7702	2999.3691 / 3347.3964	3000.6831 / 3349.4392	3002.3742 / 3347.4267	3000.6008 / 3347.0729	3000.3935 / 3348.7676	3001.7728 / 3346.4543
	TIP-IM (IJCAI 2021)	1368.2479 / 1759.2384	1344.0287 / 1746.3587	1354.7728 / 1758.6607	1354.3781 / 1758.4028	1377.2661 / 1763.0295	1332.3683 / 1744.8739	1355.3942 / 1738.2468
	Clip2Protect (CVPR 2023)	11970.8025 / 6859.8500	11924.3039 / 6796.8330	11966.5012 / 6838.9705	11969.2163 / 6799.9694	11980.1491 / 6871.7891	11928.2899 / 6856.1796	11908.1188 / 6864.5540
	Sibling-Attack (CVPR 2023)	2635.2834 / 2825.9886	2635.2834 / 2825.9886	2635.2834 / 2825.9886	2635.2834 / 2825.9886	2635.2834 / 2825.9886	2635.2834 / 2825.9886	2635.2834 / 2825.9886
	ATT (NIPS 2024)	2349.0473 / 2471.6685	2349.0473 / 2471.6685	2349.0473 / 2471.6685	2349.0473 / 2471.6685	2349.0473 / 2471.6685	2349.0473 / 2471.6685	2349.0473 / 2471.6685
	AdvDiffusion (AAAI 2024)	10655.6053 / 10092.4532	10577.9282 / 10095.6835	10620.8835 / 10127.6337	10541.8796 / 10116.3258	10649.9894 / 10055.1768	10592.9780 / 10165.5424	10346.1927 / 10164.8608
	DPA (CVPR 2025)	1609.2033 / 1825.5179	1609.2033 / 1825.5179	1609.2033 / 1825.5179	1609.2033 / 1825.5179	1609.2033 / 1825.5179	1609.2033 / 1825.5179	1609.2033 / 1825.5179
	PPOM-ATTACK (Ours)	1126.1346 / 1167.8196	1076.2591 / 1089.5181	1035.8449 / 1018.5211	863.1992 / 1016.7477	1051.9264 / 1867.6077	985.5096 / 1225.9675	1437.6681 / 1811.9060
LPIPS(↓)	PGD	0.0507 / 0.0604	0.0453 / 0.0581	0.0496 / 0.0598	0.0491 / 0.0596	0.0547 / 0.0598	0.0460 / 0.0588	0.0528 / 0.0635
	TI-RDIM	0.1838 / 0.1159	0.1864 / 0.1171	0.1833 / 0.1151	0.1840 / 0.1152	0.1855 / 0.1164	0.1831 / 0.1139	0.1824 / 0.1156
	TIP-IM (IJCAI 2021)	0.0974 / 0.0839	0.0920 / 0.0820	0.0964 / 0.0842	0.0964 / 0.0841	0.0924 / 0.0824	0.0934 / 0.0823	0.0934 / 0.0830
	Clip2Protect (CVPR 2023)	0.3425 / 0.1737	0.3425 / 0.1733	0.3423 / 0.1731	0.3427 / 0.1725	0.3447 / 0.1751	0.3425 / 0.1747	0.3417 / 0.1728
	Sibling-Attack (CVPR 2023)	0.1720 / 0.1298	0.1720 / 0.1298	0.1720 / 0.1298	0.1720 / 0.1298	0.1720 / 0.1298	0.1720 / 0.1298	0.1720 / 0.1298
	ATT (NIPS 2024)	0.2129 / 0.1539	0.2129 / 0.1539	0.2129 / 0.1539	0.2129 / 0.1539	0.2129 / 0.1539	0.2129 / 0.1539	0.2129 / 0.1539
	AdvDiffusion (AAAI 2024)	0.3053 / 0.2411	0.3040 / 0.2397	0.3052 / 0.2422	0.3040 / 0.2422	0.3056 / 0.2378	0.3042 / 0.2399	0.2996 / 0.2398
	DPA (CVPR 2025)	0.1164 / 0.0822	0.1164 / 0.0822	0.1164 / 0.0822	0.1164 / 0.0822	0.1164 / 0.0822	0.1164 / 0.0822	0.1164 / 0.0822
	PPOM-ATTACK (Ours)	0.0062 / 0.0030	0.0057 / 0.0026	0.0035 / 0.0023	0.0039 / 0.0025	0.0062 / 0.0094	0.0051 / 0.0036	0.0129 / 0.0072
DISTS(↓)	PGD	0.1313 / 0.1250	0.1258 / 0.1202	0.1300 / 0.1236	0.1281 / 0.1231	0.1391 / 0.1247	0.1225 / 0.1201	0.1339 / 0.1311
	TI-RDIM	0.2454 / 0.1949	0.2470 / 0.1953	0.2454 / 0.1943	0.2456 / 0.1930	0.2451 / 0.1950	0.2442 / 0.1926	0.2425 / 0.1927
	TIP-IM (IJCAI 2021)	0.1560 / 0.1321	0.1526 / 0.1296	0.1582 / 0.1320	0.1583 / 0.1323	0.1510 / 0.1319	0.1540 / 0.1299	0.1521 / 0.1311
	Clip2Protect (CVPR 2023)	0.3027 / 0.1930	0.3019 / 0.1924	0.3024 / 0.1926	0.3021 / 0.1918	0.3041 / 0.1940	0.3020 / 0.1937	0.3014 / 0.1921
	Sibling-Attack (CVPR 2023)	0.2134 / 0.1936	0.2134 / 0.1936	0.2134 / 0.1936	0.2134 / 0.1936	0.2134 / 0.1936	0.2134 / 0.1936	0.2134 / 0.1936
	ATT (NIPS 2024)	0.2066 / 0.1886	0.2066 / 0.1886	0.2066 / 0.1886	0.2066 / 0.1886	0.2066 / 0.1886	0.2066 / 0.1886	0.2066 / 0.1886
	AdvDiffusion (AAAI 2024)	0.2815 / 0.2356	0.2806 / 0.2362	0.2809 / 0.2370	0.2808 / 0.2372	0.2816 / 0.2345	0.2812 / 0.2366	0.2769 / 0.2380
	DPA (CVPR 2025)	0.2018 / 0.1931	0.2018 / 0.1931	0.2018 / 0.1931	0.2018 / 0.1931	0.2018 / 0.1931	0.2018 / 0.1931	0.2018 / 0.1931
	PPOM-ATTACK (Ours)	0.0514 / 0.0460	0.0494 / 0.0392	0.0360 / 0.0400	0.0379 / 0.0394	0.0517 / 0.0815	0.0458 / 0.0524	0.0740 / 0.0682

an average of 21.7% in terms of ASR. Similarly, on the CelebA-HQ dataset, PPOM-ATTACK demonstrates better performance, achieving an average improvement of 22.9% in ASR. The experimental results indicate that our method can enhance the ASR without using substitute models but directly predict and optimize the perturbation region according to the feedback from the target FR model.

- From the experimental results from Table II, PPOM-ATTACK achieves higher PSNR and SSIM than those of the most SOTA methods on the LFW and CelebA-HQ datasets. The visualization of adversarial face images generated by each attack methods are shown in Figure 4. The results demonstrate that PPOM-ATTACK preserves the clarity and remains free of perceptible artifacts in adversarial face images. In contrast, other SOTA methods, with the exception of PGD, introduce obvious artifacts. This indicates that utilizing minimum brightness offset perturbations can effectively maintain the image quality of adversarial face images.
- The experimental results from Table II shows that PPOM-ATTACK achieves lower l_2 norm than that of the most SOTA methods on the LFW and CelebA-HQ datasets, which means PPOM-ATTACK take less perturbation magnitude compared with most SOTA methods.
- In addition, shuffleFace uses DenseNet blocks, Center-loss employs VGGFace as its backbone, whereas the other methods utilize ResNet as their backbone architecture. The different network structures make an obvious

difference of substitute models that leads to a decrease of ASR observed in methods that rely on substitute models.

- Beyond traditional pixel-level metrics, we further evaluate perceptual similarity using LPIPS and DISTS on both LFW and CelebA-HQ datasets. As shown in Tables II, PPOM-ATTACK achieves the lowest LPIPS and DISTS values across all target FR models, significantly outperforming existing SOTA methods. On the LFW dataset, PPOM-ATTACK reduces LPIPS and DISTS by large margins compared with diffusion-based and surrogate-model-based approaches, indicating superior preservation of perceptual similarity and structural texture fidelity in adversarial images. Similar trends persist on the CelebA-HQ dataset, where PPOM-ATTACK achieves LPIPS scores close to 0.002–0.009 and consistently lower DISTS values than competing methods. These results show that our perturbation prediction and optimization strategy can maintain high attack success rates while introducing smaller perceptual deviations than diffusion-based and substitute model driven baselines, which exhibit higher LPIPS/DISTS values and more pronounced artifacts.

3) *Evaluation on Attacking Commercial API:* To further demonstrate the effectiveness of our PPOM-ATTACK in real-world scenarios, we conduct comparison experiments on Face++ API. To align with real-world scenarios, we use



Fig. 4. Comparison of adversarial face images on CelebA-HQ dataset generated by different methods for different target models.

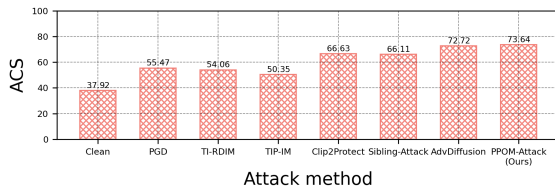


Fig. 5. Average confidence score (ACS, higher is better) for impersonation attack returned by Face++ API.

MTCNN [45] to extract the face region and apply a mask to cover the non-face regions, ensuring that the non-face regions are not selected by the agent. Additionally, to obtain more pronounced feedback, we set the single-step brightness offset to ± 5 .

We compare average confidence score from Face++ API of our PPOM-ATTACK method with SOTA methods on the CelebA-HQ dataset, as shown in Figure 5. The results shows that our method outperforms SOTA methods on Face++ API in terms of ASR.

Additionally, the visualization and query results (confidence score from Face++ API) of adversarial face images generated by each attack method is shown in Figure 6.

4) *Evaluation of Robust FR Models With Defence Strategies*: We further compare the ASR of our PPOM-Attack

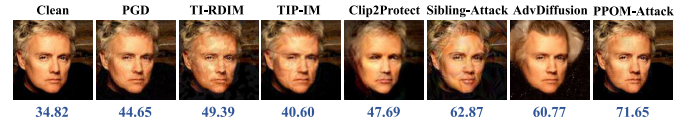


Fig. 6. Visualization of our adversarial face image comparing with other attacks. Each column shows the adversarial face image and the query result (confidence score, blue) from Face++.

TABLE III

ASR RESULTS OF BLACK-BOX IMPERSONATION ATTACK ON ADVERSARIAL TRAINED DEFENSE MODELS. WE CHOOSE METHODS EXHIBITING STRONGER TRANSFERABILITY IN THE MANUSCRIPTS (SIBLING-ATTACK, CLIP2PROTECT AND ADVDIFFUSION) FOR COMPARISONS OF OUR PPOM-ATTACK METHODS. AT REPRESENTS PGD-AT

Attack method / FR model	AT		TRADES	
	LFW / CelebA-HQ		LFW / CelebA-HQ	
Sibling-Attack	2.8	14.4	4.6	43.2
Clip2Protect	4.8	15.0	22.0	64.4
AdvDiffusion	3.8	16.8	12.6	60.4
PPOM-ATTACK(Ours)	20.6	36.8	54.8	92.4

method with performance-stronger attack methods such as Clip2Protect, Sibling-Attack and AdvDiffusion targeting robust FR models with adversarial trained defence strategy. Since adversarially trained models have been shown to be an effective defense against adversarial examples [1], we evaluate the ASR on two adversarial trained FR models: PGD-AT [1] and TRADES [43]. The thresholds α for computing the ASR of these two FR models will be set as 0.8107 and 0.3707 for PGD-AT, TRADES, respectively.

As shown in Table III, the results show that our method outperforms the best competitors by 15.8% (PGD-AT) and 32.8% (TRADES) on LFW as well as 20.0% and 28.0% on CelebA-HQ, which means our PPOM-ATTACK method is more efficiency than the SOTA methods against adversarial trained defence strategy.

5) *Evaluation of Query-Based Attack Methods*: We compare PPOM-ATTACK with representative query-based adversarial attack methods on the LFW and CelebA-HQ datasets, as summarized in Tables IV. Across all FR models on both datasets, PPOM-ATTACK attains the highest SSIM and PSNR and the lowest l_2 , LPIPS, and DISTS values among all methods, indicating that the required perturbations are smaller in magnitude and remain closer to the original images in both pixel-level and learned perceptual feature space. The visualization of adversarial face images generated by each attack methods are shown in Figure 4.

Boundary-based methods such as Evolution, CGBA and DevoPatch frequently achieve 100% ASR under the 10k query budget for all FR models. This behavior is consistent with their boundary-based optimization strategy: instead of starting from the clean source image, they first search for an adversarial image that already lies on the target side of the decision boundary, i.e., satisfies $D(f(x^d), f(x^t)) \leq \alpha$. Once such a successful adversarial example has been found for a given pair, subsequent queries are mainly used to reduce the distortion while keeping the example misclassified. This success-first, distortion-reduction procedure naturally keeps the final ASR

TABLE IV
PERFORMANCE COMPARISON WITH SOTA QUERY-BASED METHODS ON LFW AND CELEBA-HQ DATASETS

Metric	Attack method / FR model	IR50	MobileFace	CurricularFace	InsightFace	ShuffleFace	Arcface	Centerloss
		LFW / CelebA-HQ	LFW / CelebA-HQ	LFW / CelebA-HQ	LFW / CelebA-HQ	LFW / CelebA-HQ	LFW / CelebA-HQ	LFW / CelebA-HQ
ASR(%)	Sparse-RS (AAAI 2022)	86.0 / 67.0	94.0 / 79.4	92.8 / 94.2	98.0 / 87.4	98.0 / 22.8	84.6 / 78.8	47.0 / 60.4
	BruSLeAttack (ICLR 2024)	90.0 / 73.0	98.0 / 77.4	94.8 / 94.2	100.0 / 91.4	100.0 / 24.8	88.6 / 82.8	51.0 / 60.4
	GSBAK (ICLR 2025)	42.0 / 44.0	52.0 / 54.0	92.8 / 92.2	86.4 / 86.0	32.0 / 4.0	88.6 / 82.4	28.0 / 30.0
	Evolution (CVPR 2019)	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0
	CGBA (ICCV 2023)	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0
	DevoPatch (TIFS 2023)	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0
	PPOM-ATTACK (Ours)	100.0 / 87.0	100.0 / 89.4	94.8 / 96.2	100.0 / 91.4	99.6 / 64.8	94.6 / 96.8	67.0 / 70.4
SSIM(↑)	Sparse-RS (AAAI 2022)	0.8789 / 0.9132	0.8781 / 0.9083	0.8801 / 0.9126	0.8807 / 0.9125	0.8791 / 0.9155	0.8773 / 0.9147	0.8774 / 0.9107
	BruSLeAttack (ICLR 2024)	0.8700 / 0.9065	0.8677 / 0.9030	0.8701 / 0.9044	0.8712 / 0.9032	0.8703 / 0.9089	0.8698 / 0.9084	0.8705 / 0.9027
	GSBAK (ICLR 2025)	0.8810 / 0.8885	0.8942 / 0.8949	0.9134 / 0.9107	0.9076 / 0.9037	0.8755 / 0.8726	0.9074 / 0.8932	0.8794 / 0.8813
	Evolution (CVPR 2019)	0.6732 / 0.7002	0.7783 / 0.8124	0.7330 / 0.7437	0.7431 / 0.7390	0.6933 / 0.6402	0.7391 / 0.7148	0.6349 / 0.6355
	CGBA (ICCV 2023)	0.6203 / 0.5970	0.7392 / 0.7522	0.7011 / 0.6629	0.7092 / 0.6661	0.6188 / 0.5198	0.6880 / 0.6088	0.5079 / 0.4375
	DevoPatch (TIFS 2023)	0.8690 / 0.9046	0.9186 / 0.9043	0.8893 / 0.9103	0.8946 / 0.9157	0.8520 / 0.8373	0.8937 / 0.9109	0.7601 / 0.8097
	PPOM-ATTACK (Ours)	0.9616 / 0.9717	0.9643 / 0.9726	0.9758 / 0.9764	0.9743 / 0.9761	0.9640 / 0.9381	0.9681 / 0.9684	0.9368 / 0.9505
PSNR(↑)	Sparse-RS (AAAI 2022)	27.4620 / 28.4468	27.4513 / 28.3982	27.5810 / 28.5508	27.5573 / 28.5811	27.7355 / 28.4314	27.5400 / 28.5264	27.4157 / 28.2105
	BruSLeAttack (ICLR 2024)	26.8429 / 27.6796	26.8744 / 27.6675	26.8689 / 27.6807	26.7933 / 27.6040	26.9628 / 27.8532	26.9598 / 27.8150	26.8689 / 27.4740
	GSBAK (ICLR 2025)	32.0266 / 32.7068	32.8687 / 33.5274	34.2667 / 34.1645	33.8388 / 33.4420	31.7622 / 31.6556	33.8129 / 33.0910	32.2473 / 32.2738
	Evolution (CVPR 2019)	17.5300 / 17.5462	20.4819 / 21.1608	19.4403 / 18.5469	19.2965 / 18.4203	17.9945 / 15.9512	19.7619 / 17.7736	17.4999 / 15.9955
	CGBA (ICCV 2023)	16.6305 / 15.9923	20.1039 / 20.5044	18.6615 / 18.8226	18.4458 / 17.1195	16.9691 / 14.0578	18.1252 / 15.4937	15.6329 / 13.4183
	DevoPatch (TIFS 2023)	23.1167 / 23.5781	26.2500 / 28.7096	24.1412 / 25.5896	24.5388 / 25.9681	21.9333 / 20.2191	24.0568 / 23.6565	19.6185 / 20.3275
	PPOM-ATTACK (Ours)	33.2540 / 34.2912	33.8222 / 35.2824	35.9003 / 35.4276	35.7524 / 35.4713	33.9150 / 29.7833	34.5162 / 33.6735	31.3367 / 31.5442
$l_2(\downarrow)$	Sparse-RS (AAAI 2022)	2109.8116 / 2151.7065	2112.7562 / 2165.4098	2078.8008 / 2126.0597	2085.9823 / 2119.7776	2046.2470 / 2155.7673	2092.9423 / 2132.3716	2119.0856 / 2213.0120
	BruSLeAttack (ICLR 2024)	2266.5118 / 2348.0901	2256.2506 / 2352.8604	2257.8083 / 2349.0584	2275.9714 / 2371.7326	2232.5877 / 2304.5935	2238.2990 / 2316.4374	2257.3886 / 2410.5427
	GSBAK (ICLR 2025)	1248.6055 / 1351.4650	1160.1556 / 1279.8265	1016.0147 / 1174.3992	1056.4754 / 1236.1397	1280.3533 / 1478.3023	1069.5626 / 1311.0154	1251.7158 / 1403.6125
	Evolution (CVPR 2019)	6714.1057 / 7834.2330	5027.3968 / 5470.5548	5740.7066 / 6891.0839	5580.3273 / 6986.0069	6416.6693 / 9124.1217	5678.6519 / 7542.2058	7261.4770 / 9157.7670
	CGBA (ICCV 2023)	7509.2188 / 9543.1142	5390.3386 / 6246.9974	6380.2310 / 8465.0066	6184.8925 / 8238.3854	7335.2932 / 11461.2669	6609.4525 / 9975.1364	8780.0999 / 12678.9968
	DevoPatch (TIFS 2023)	3699.0337 / 4169.9842	2767.0411 / 2710.1698	3426.6304 / 3352.3308	3268.5888 / 3219.7861	4226.7646 / 5797.0228	3384.7766 / 3966.7267	5507.0106 / 6100.3489
	PPOM-ATTACK (Ours)	1126.1346 / 1167.8196	1076.6291 / 1089.5181	835.8449 / 1018.5211	863.1992 / 1016.7477	1051.9264 / 1867.6077	985.5096 / 1225.9675	1437.6681 / 1611.9060
LPIPS(↓)	Sparse-RS (AAAI 2022)	0.1720 / 0.0921	0.1711 / 0.0966	0.1647 / 0.0865	0.1613 / 0.0890	0.1552 / 0.0932	0.1757 / 0.0910	0.1701 / 0.0949
	BruSLeAttack (ICLR 2024)	0.1932 / 0.1116	0.1916 / 0.1124	0.1891 / 0.1105	0.1812 / 0.1104	0.1778 / 0.1085	0.1948 / 0.1116	0.1888 / 0.1147
	GSBAK (ICLR 2025)	0.0740 / 0.0333	0.0645 / 0.0307	0.0519 / 0.0250	0.0563 / 0.0263	0.0792 / 0.0389	0.0528 / 0.0297	0.0756 / 0.0345
	Evolution (CVPR 2019)	0.1419 / 0.1184	0.0928 / 0.0574	0.1132 / 0.0993	0.1077 / 0.0949	0.1307 / 0.1552	0.1147 / 0.1042	0.1714 / 0.1487
	CGBA (ICCV 2023)	0.2051 / 0.1922	0.1352 / 0.0959	0.1465 / 0.1529	0.1397 / 0.1463	0.2095 / 0.2363	0.1520 / 0.1787	0.2922 / 0.2924
	DevoPatch (TIFS 2023)	0.1026 / 0.0761	0.0753 / 0.0361	0.0842 / 0.0445	0.0802 / 0.0423	0.1330 / 0.1075	0.0988 / 0.0541	0.1519 / 0.1122
	PPOM-ATTACK (Ours)	0.0062 / 0.0030	0.0057 / 0.0026	0.0035 / 0.0023	0.0039 / 0.0025	0.0062 / 0.0094	0.0051 / 0.0036	0.0129 / 0.0072
DISTS(↓)	Sparse-RS (AAAI 2022)	0.1762 / 0.1370	0.1772 / 0.1473	0.1800 / 0.1416	0.1766 / 0.1418	0.1757 / 0.1391	0.1780 / 0.1431	0.1768 / 0.1432
	BruSLeAttack (ICLR 2024)	0.1876 / 0.1497	0.1867 / 0.1559	0.1889 / 0.1515	0.1863 / 0.1522	0.1827 / 0.1463	0.1869 / 0.1505	0.1842 / 0.1517
	GSBAK (ICLR 2025)	0.1388 / 0.1118	0.1283 / 0.1042	0.1107 / 0.0968	0.1126 / 0.0979	0.1434 / 0.1216	0.1122 / 0.1048	0.1366 / 0.1147
	Evolution (CVPR 2019)	0.1811 / 0.1657	0.1482 / 0.1130	0.1623 / 0.1538	0.1584 / 0.1499	0.1742 / 0.1893	0.1632 / 0.1588	0.1960 / 0.1837
	CGBA (ICCV 2023)	0.2154 / 0.2148	0.1741 / 0.1548	0.1812 / 0.1917	0.1785 / 0.1873	0.2174 / 0.2325	0.1814 / 0.1991	0.2616 / 0.2641
	DevoPatch (TIFS 2023)	0.1026 / 0.0761	0.0710 / 0.0512	0.0883 / 0.0638	0.0983 / 0.0573	0.1144 / 0.1129	0.0870 / 0.0647	0.1519 / 0.1210
	PPOM-ATTACK (Ours)	0.0514 / 0.0460	0.0494 / 0.0392	0.0360 / 0.0400	0.0379 / 0.0394	0.0517 / 0.0815	0.0458 / 0.0524	0.0740 / 0.0682
Avg. queries (↓)	Sparse-RS (AAAI 2022)	2874 / 3929	1389 / 2912	1308 / 1552	944 / 1691	1401 / 7607	2767 / 4525	5148 / 5391
	BruSLeAttack (ICLR 2024)	2775 / 3776	1238 / 3073	1294 / 1390	803 / 1801	1025 / 7489	2500 / 4583	5073 / 5258
	GSBAK (ICLR 2025)	6438 / 6182	5511 / 5157	1272 / 1275	1961 / 2431	7357 / 9742	1854 / 2946	7533 / 7425
	Evolution (CVPR 2019)	10000 / 10000	10000 / 10000	10000 / 10000	10000 / 10000	10000 / 10000	10000 / 10000	10000 / 10000
	CGBA (ICCV 2023)	10172 / 10175	10164 / 10146	10166 / 10180	10190 / 10173	10155 / 10150	10190 / 10153	10168 / 10144
	DevoPatch (TIFS 2023)	10000 / 10000	10000 / 10000	10000 / 10000	10000 / 10000	10000 / 10000	10000 / 10000	10000 / 10000
	PPOM-ATTACK (Ours)	2007 / 1968	1638 / 1359	1216 / 1484	1252 / 1553	2161 / 6150	1635 / 2053	4869 / 4886

TABLE V
ABLATION EXPERIMENTS FOR PPOM-ATTACK ON LFW DATASET

Metric	Attack method / FR model	IR50	MobileFace	CurricularFace	InsightFace	ShuffleFace	Arcface	Centerloss
ASR(%)	Full-sampling + Block-offset	98.0	98.0	94.8	100.0	97.6	94.6	67.0
	PPO-prediction + Pixel-offset	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	PPO-prediction + Block-offset (Ours)	100.0	100.0	94.8	100.0	99.6	94.6	67.0
SSIM(↑)	Full-sampling + Block-offset	0.9251	0.9275	0.9361	0.9316	0.9335	0.9294	0.9023
	PPO-prediction + Pixel-offset	0.9994	0.9996	0.9990	0.9993	0.9988	0.9989	0.9990
	PPO-prediction + Block-offset (Ours)	0.9616	0.9643	0.9758	0.9743	0.9640	0.9681	0.9368
PSNR(↑)	Full-sampling + Block-offset	30.5564	30.9728	31.8962	31.7004	31.4552	31.4502	29.6291
	PPO-prediction + Pixel-offset	57.0529	59.4976	54.7811	56.3568	53.8954	54.4026	54.6282
	PPO-prediction + Block-offset (Ours)	33.2540	33.8222	35.9093	35.7524	33.9150	34.5162	31.3367
$l_2(\downarrow)$	Full-sampling + Block-offset	1478.8951	1437.3396	1298.8210	1311.2753	1340.8872	1366.0822	1690.5648
	PPO-prediction + Pixel-offset	69.4994	53.1466	90.2387	75.3354	100.0775	94.3971	91.9760
	PPO-prediction + Block-offset (Ours)	1126.1346	1076.2591	835.8449	863.1992	1051.9264	985.5096	1437.6681
Avg. queries(↓)	Full-sampling + Block-offset	13122	11906	10003	10373	11768	11358	17753
	PPO-prediction + Pixel-offset	9900	9900	9900	9900	9900	9900	9900
	PPO-prediction + Block-offset (Ours)	2007	1638	1216	1252	2161	1635	4869

at 100% within the allowed query budget. In our experiments, however, Evolution and CGBA still fail to converge to a low perturbation magnitude even within a 10k query budget, producing significantly larger modifications than other methods, as reflected by their lower PSNR/SSIM and higher l_2 , LPIPS, and DISTS values.

In contrast, PPOM-ATTACK, Sparse-RS, BruSLeAttack, and GSBK all starts from the original image x^0 and directly optimizes perturbations to cross the decision boundary from the non-target side, aiming to achieve $D(f(x^a), f(x^t)) \leq \alpha$ while operating under a capped query budget. Because no large adversarial initialization is used and the number of queries is limited, some difficult pairs do not cross the boundary, so the ASR remains slightly below 100% for several FR models. Compared with Sparse-RS and BruSLeAttack, PPOM-ATTACK achieves comparable or higher ASR on most FR models while using fewer queries on CelebA-HQ

and a similar query budget on LFW, and at the same time consistently yields better image quality. Compared with GSBK, PPOM-ATTACK attains substantially higher ASR on LFW (and competitive ASR on CelebA-HQ) while maintaining markedly lower perceptual distortion (LPIPS/DISTS) and higher structural similarity. We note that GSBK can obtain higher PSNR or lower l_2 on a few specific FR models, but this advantage is accompanied by a clear degradation in attack success rate under the same query constraints.

6) *Ablation Experiments*: To show the necessity of individual components in our proposed PPOM-ATTACK framework, including PPO-prediction and Block-offset, we conducted ablation experiments by comparing three configurations: Full-sampling with Block-offset, PPO-prediction with Pixel-offset, and PPO-prediction with Block-offset (PPOM-ATTACK). As shown in Table V, VI, the Pixel-offset baseline struggles significantly, with ASR values mostly limited to 1%, revealing

TABLE VI
ABLATION EXPERIMENTS FOR PPOM-ATTACK ON CELEBA-HQ DATASET

Metric	Attack method / FR model	IR50	MobileFace	CurricularFace	InsightFace	ShuffleFace	Arcface	Centerloss
ASR(%)	Full-sampling + Block-offset	87.0	89.4	96.2	91.4	28.8	96.8	57.4
	PPO-prediction + Pixel-offset	1.2	8.8	1.0	1.8	0.2	0.4	1.0
	PPO-prediction + Block-offset (Ours)	87.0	89.4	96.2	91.4	64.8	96.8	70.4
SSIM(↑)	Full-sampling + Block-offset	0.9319	0.9360	0.9363	0.9321	0.8854	0.9228	0.9032
	PPO-prediction + Pixel-offset	0.9998	0.9999	0.9996	0.9998	0.9994	0.9995	0.9995
	PPO-prediction + Block-offset (Ours)	0.9717	0.9726	0.9764	0.9761	0.9381	0.9684	0.9505
PSNR(↑)	Full-sampling + Block-offset	30.6368	31.1877	31.2162	31.0324	28.5260	30.8477	29.3519
	PPO-prediction + Pixel-offset	64.4663	71.8264	60.2968	62.7065	58.5228	58.5079	59.8612
	PPO-prediction + Block-offset (Ours)	34.2912	35.2824	35.4276	35.4713	29.7833	33.6735	31.5442
$l_2(\downarrow)$	Full-sampling + Block-offset	1682.8126	1596.7435	1573.4167	1600.7673	2132.8298	1691.5879	1960.9532
	PPO-prediction + Pixel-offset	34.6313	15.6552	54.9193	41.6222	67.2417	67.2972	58.4607
	PPO-prediction + Block-offset (Ours)	1167.8196	1089.5181	1018.5211	1016.7477	1867.6077	1225.9675	1611.9060
Avg. queries(↓)	Full-sampling + Block-offset	17100	13700	14932	14757	28359	17502	24034
	PPO-prediction + Pixel-offset	9880	9120	9900	9820	9980	9960	9900
	PPO-prediction + Block-offset (Ours)	1968	1359	1484	1553	6150	2053	4886

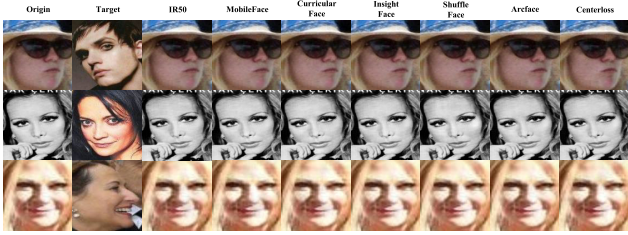


Fig. 7. The adversarial face images on VGGFace2 dataset generated by our method for different target models.

TABLE VII
PERFORMANCE OF PPOM-ATTACK ON VGGFACE2 ACROSS DIFFERENT FR MODELS

Model	ASR(%)	SSIM	PSNR	l_2	Avg. queries
IR50	100.0	0.9720	34.42	1138.86	1684
MobileFace	100.0	0.9810	37.54	887.02	803
CurricularFace	100.0	0.9828	37.20	851.12	944
InsightFace	100.0	0.9834	37.77	819.19	922
ShuffleFace	97.8	0.9652	34.36	1232.16	2266
ArcFace	100.0	0.9812	37.06	874.62	1074
Centerloss	87.8	0.9540	32.20	1485.52	3789

its inefficiency in enabling successful attacks. Meanwhile, the Full-sampling configuration suffers from much higher query requirements.

C. Evaluation of Challenging Face Datasets

To further assess the behavior of PPOM-ATTACK under more diverse and challenging conditions, we evaluate the proposed method on the VGGFace2 dataset, which contains large variations in pose, illumination, expression, and occlusion. Table VII reports the results across all seven target FR models. And the visualization of adversarial face images generated by our method on VGGFace2 are shown in Figure 7.

On VGGFace2, PPOM-ATTACK attains ASR values of 100% for IR50, MobileFace, CurricularFace, InsightFace, and ArcFace, and 97.8% and 87.8% for ShuffleFace and Centerloss, respectively. The SSIM scores lie between 0.9540 and 0.9834, and the PSNR values range from 32.20 dB to 37.77 dB, indicating that the adversarial images remain close to the original images in terms of pixel-level similarity. The l_2 perturbation norms are between approximately 8.2×10^2 and 1.5×10^3 across all models. The average number of queries

per image is below 2,300 for six FR models (ranging from 803 to 2,266) and 3,789 for Centerloss.

Overall, these results indicate that PPOM-ATTACK achieves high ASR and moderate perturbation magnitudes on VGGFace2, suggesting that the method extends to a dataset with substantial variations in pose, lighting, and occlusion.

D. Parameter Analysis

We conduct a parameter study to investigate the impact of the perturbation region size ρ on ASR, image quality, and query efficiency, and to select a default value for our method. Table VIII reports results for $\rho \in \{0.002, 0.02, 0.2\}$ on both LFW and CelebA-HQ across all target FR models.

- Small regions ($\rho = 0.002$). When ρ is very small, only a small fraction of pixels is modified in each step. For all FR models on both datasets, this setting yields the highest SSIM and PSNR values, indicating that the perturbations remain closest to the original images in terms of pixel-level similarity. However, the attack success rates (ASR) are substantially lower than those for larger ρ , and the average number of queries is the largest in all cases. This shows that, within a limited query budget, smaller localized perturbations are insufficient to reliably push adversarial examples across the decision boundary required for a successful attack.
- Large regions ($\rho = 0.2$). When ρ is large, the perturbation region covers a much larger portion of the face. This increases the effect of a single brightness offset and generally leads to higher ASR than $\rho = 0.002$, with ASR values remaining high across most FR models on both datasets. At the same time, SSIM and PSNR decrease monotonically as ρ increases from 0.002 to 0.2, and $\rho = 0.2$ consistently produces the lowest SSIM and PSNR among the three settings, indicating a stronger deviation from the original images. The average query counts for $\rho = 0.2$ are lower than those for $\rho = 0.002$ and are comparable to or slightly lower than those for $\rho = 0.02$ for several models, reflecting more aggressive but visually more noticeable perturbations.
- Intermediate regions ($\rho = 0.02$). The intermediate value $\rho = 0.02$ provides a compromise between these two extremes. Across LFW and CelebA-HQ, $\rho = 0.02$ achieves ASR values that are close to, and in several cases

TABLE VIII
IMPACT OF ρ ON OUR METHOD ON LFW AND CELEBA-HQ

Model	ASR(%)			SSIM($\uparrow$)			PSNR($\uparrow$)			Avg. queries		
	$\rho=0.002$	$\rho=0.02$	$\rho=0.2$	$\rho=0.002$	$\rho=0.02$	$\rho=0.2$	$\rho=0.002$	$\rho=0.02$	$\rho=0.2$	$\rho=0.002$	$\rho=0.02$	$\rho=0.2$
IR50	55.2/80.6	100.0/87.0	91.8/92.6	0.9789/0.9881	0.9660/0.9717	0.9491/0.9507	40.1117/43.2706	33.2540/34.2912	29.4617/30.2702	4366/4652	2007/1968	1914/1623
MobileFace	53.8/79.2	100.0/89.4	89.2/92.6	0.9800/0.9870	0.9684/0.9726	0.9495/0.9530	40.4224/43.6310	33.8222/35.2824	29.4939/30.8110	3668/3563	1638/1359	1673/1226
CurricularFace	55.8/91.0	94.8/96.2	96.6/97.0	0.9861/0.9908	0.9722/0.9764	0.9611/0.9573	42.1533/44.2917	35.9093/35.4276	30.9046/31.0846	2553/3437	1216/1484	851/906
InsightFace	56.8/86.8	100.0/91.4	95.4/94.6	0.9859/0.9910	0.9721/0.9761	0.9601/0.9568	42.1127/44.3916	35.7524/35.4713	30.8244/31.1386	2650/3586	1252/1553	911/963
ShuffleFace	42.4/33.4	99.6/64.8	80.4/41.4	0.9804/0.9723	0.9255/0.9381	0.9510/0.9200	40.6319/38.4171	33.9150/29.7833	29.6400/28.0670	4415/8927	2161/6150	2283/7296
ArcFace	55.0/89.2	94.6/96.8	96.6/95.4	0.9816/0.9855	0.9628/0.9684	0.9525/0.9461	40.9045/42.1605	34.5162/33.6735	30.0626/29.8200	3415/4628	1635/2053	1498/1795
Centerloss	29.4/45.4	67.0/70.4	55.6/61.0	0.9660/0.9764	0.9404/0.9505	0.9329/0.9334	37.9563/39.9385	31.3367/31.5442	28.7088/28.9420	7172/7252	4869/4886	5635/5193

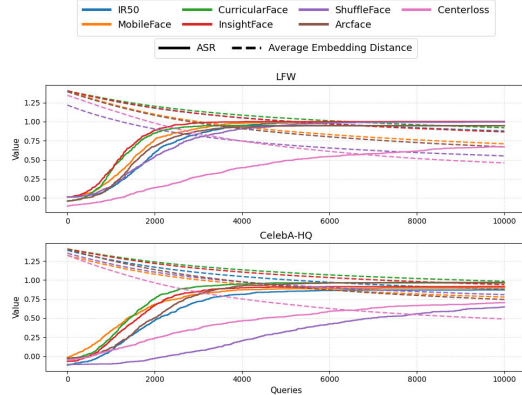


Fig. 8. The behaviors of our method from 0k to 10k queries on LFW dataset and CelebA-HQ dataset.

TABLE IX
PERFORMANCE OF PPOM-ATTACK UNDER STRICTER QUERY LIMITS ON LFW (L) AND CELEBA-HQ (C)

ASR(%) (L / C)	IR50	MobileFace	CurricularFace	InsightFace	ShuffleFace	ArcFace	Centerloss
2k queries	59.0/47.2	76.2/68.0	84.8/75.6	88.8/67.4	54.0/8.0	67.6/49.2	13.4/23.6
3k queries	84.4/71.8	91.2/78.4	94.0/91.2	98.6/81.6	79.2/18.0	87.8/80.2	28.0/35.6

higher than, those for $\rho = 0.2$, while maintaining SSIM and PSNR consistently between the $\rho = 0.002$ and $\rho = 0.2$ settings. In addition, the average number of queries for $\rho = 0.02$ is significantly smaller than for $\rho = 0.002$ in all FR models and is of the same order as that for $\rho = 0.2$. Based on these observations, we adopt $\rho = 0.02$ as the default region size in PPOM-ATTACK, as it yields high ASR with moderate degradation in SSIM/PSNR and avoids the large query counts observed for $\rho = 0.002$.

E. Evaluation Across Varying Queries

To evaluate PPOM-ATTACK under stricter query budgets, we plot the average embedding distance and ASR as functions of the number of queries from 0 to 10k. As shown in Fig. 8, the embedding distance for PPOM-ATTACK decreases rapidly within the first 1k queries, while the ASR remains low in this range because most samples have not yet reached the decision threshold used to define a successful attack. As the number of queries increases beyond 1k, the distance continues to decrease more gradually, and the ASR starts to rise once the distance falls below the threshold for a growing fraction of samples. The curves are plotted up to 10k queries, which we treat as the full-budget setting. We additionally consider fixed query caps of 2k and 3k queries per image. As shown in Table IX, under a 3k query limit, PPOM-ATTACK attains ASR values between 84.4% and 98.6% on LFW and between 71.8% and 91.2% on CelebA-HQ for IR50, MobileFace, CurricularFace,

TABLE X
IMPACT OF IMAGE RESOLUTION ON PPOM-ATTACK PERFORMANCE ON CELEBA-HQ

Model	ASR(%)			SSIM			PSNR			Avg. queries		
	128	512	1024	128	512	1024	128	512	1024	128	512	1024
IR50	87.0	88.8	88.8	0.9717	0.9723	0.9744	34.29	34.47	34.45	1968	1943	1908
MobileFace	89.4	90.6	90.2	0.9726	0.9735	0.9746	35.28	34.98	35.01	1359	1464	1483
CurricularFace	96.2	93.2	94.2	0.9764	0.9760	0.9766	35.43	35.28	35.24	1484	1520	1524
InsightFace	91.4	93.2	92.6	0.9761	0.9770	0.9781	35.47	35.45	35.50	1553	1563	1563
ShuffleFace	64.8	67.2	67.6	0.9381	0.9460	0.9512	29.78	29.81	29.87	6150	5964	5930
ArcFace	96.8	94.6	97.2	0.9684	0.9699	0.9718	33.67	33.61	33.59	2053	2084	2072
Centerloss	70.4	69.0	72.6	0.9505	0.9551	0.9594	31.54	31.59	31.70	4886	4843	4779

TABLE XI
TIME AND ENERGY BREAKDOWN OF EACH MODULE IN PPOM-ATTACK

Component	Time (s)	Energy (J)	Description
Inference (each step)	0.002	0.27	Perturbation region prediction
Training (each T step)	0.051	7.40	Perturbation region prediction
Optimal update (each step)	0.0001	0.04	Brightness offset
Query (each step)	0.0255	1.93	Brightness offset
Avg. per-step (prediction / offset)	0.0071 / 0.0256	1.01 / 1.97	$T = 10$
Total per-step	0.0327	2.97	

InsightFace, and ArcFace, while ShuffleFace and Centerloss exhibit lower ASR. When the budget is reduced to 2k queries, the ASR decreases accordingly but remains in the range of 59.0%–88.8% on LFW and 47.2%–75.6% on CelebA-HQ for the same four FR models.

F. Image Size Analysis

We conduct experiments with different image sizes to demonstrate the versatility of our method. The experimental results presented in Table X demonstrate that PPOM-ATTACK is robust to variations in image size. Across three different image sizes (128×128 , 512×512 , and 1024×1024), the ASR and image quality metrics such as SSIM and PSNR show nearly the same. This consistency in performance regardless of the image size makes PPOM-ATTACK a versatile tool for adversarial attacks against FR models.

G. Evaluation of Practical Efficiency

To evaluate the practical efficiency of PPOM-ATTACK, we conduct system level profiling and report runtime, energy consumption, and GPU memory usage on a per step basis. We sample GPU power at 10 ms intervals and integrate it over time to estimate energy consumption. As shown in Table XI, the perturbation region prediction phase introduces an inference overhead of only 0.002 s per step and consumes 0.27 J per step, while the on-policy update of the policy/value networks performed every $T = 10$ steps averages 0.051 s and 7.40 J per update (0.0071 s and 1.01 J amortized per step). The minimum brightness offset module incurs 0.0001 s and 0.04 J of local computation plus 0.0255 s and 1.93 J of query cost per

TABLE XII

CROSS-DATASET TRANSFERABILITY OF THE PPO POLICY UNDER TWO INITIALIZATION SCHEMES (SCRATCH TRAINING AND VGGFACE2 PRETRAINING), WITH ASR(%) AND AVG.Q EVALUATED ON LFW (L) AND CELEBA-HQ (C)

Model	Scratch		Transfer	
	ASR (L/C)	Avg.Q (L/C)	ASR (L/C)	Avg.Q (L/C)
IR50	100.0/87.0	2007/1968	100.0/100.0	1979/2056
MobileFace	100.0/89.4	1638/1359	100.0/100.0	1588/1478
CurricularFace	94.8/96.2	1216/1484	100.0/100.0	1187/1503
InsightFace	100.0/91.4	1252/1553	100.0/100.0	1214/1556
ShuffleFace	99.6/64.8	2161/6150	99.0/80.0	2286/6115
ArcFace	94.6/96.8	1635/2053	100.0/100.0	1568/2237
Centerloss	67.0/70.4	4869/4886	80.0/73.0	4807/4894

step, corresponding to two model evaluations per step. Overall, PPOM-ATTACK requires approximately 0.0327 s and 2.97 J per step. In terms of GPU memory footprint, we measure peak memory using PyTorch peak allocated/reserved statistics and NVML system memory usage. The PPO policy inference reaches a peak of 440.59 MB GPU allocated memory (528 MB reserved), and PPO training peaks at 475.95 MB allocated (528 MB reserved) per update. The query step (local target model) peaks at 450.95 MB allocated (528 MB reserved). Under our test setting, the NVML reported system GPU memory usage stays around 3.14 GB, with an additional transient increase during PPO updates (up to 252 MB in our runs).

Table XI further indicates that the dominant cost arises from the model query operation, which is inherent to query-based black-box attacks. Since our framework issues only two queries per step, both runtime and energy consumption scale linearly with the query budget, providing predictable and practical performance for real world scenarios.

H. Evaluation of Cross-Dataset Transfer

To examine whether the learned perturbation region strategy can be transferred across datasets (thereby reducing the cold-start overhead), we pretrain the PPO agent on VGGFace2 and directly initialize the agent for attacks on LFW and CelebA-HQ. Table XII summarizes the results. Compared with training from scratch on the target dataset, transfer pretraining increases the ASR, while the corresponding average number of queries remains of comparable magnitude across settings. This indicates that the pretrained policy serves as a better initialization and helps reduce the instability of early stage exploration under a fixed query budget.

I. Evaluation of End-to-End Training Strategy

To reduce pipeline complexity, we implement an end-to-end variant where the policy network directly outputs both components. Specifically, we extend the policy with an additional task head to predict a discrete perturbation magnitude in $\{-\Delta, 0, +\Delta\}$ for the selected region, enabling direct perturbation generation from the policy output and removing the standalone magnitude-optimization step.

Table XIII reports a comparison between the original method and the end-to-end variant on LFW and CelebA-HQ. The end-to-end variant reduces the average number of queries for most models, while its ASR varies across target models: it remains close to the original method for some models

TABLE XIII

COMPARISON BETWEEN THE ORIGINAL PPOM-ATTACK AND THE END-TO-END VARIANT. THE END-TO-END POLICY PREDICTS BOTH THE PERTURBATION REGION AND A DISCRETE PERTURBATION MAGNITUDE IN $\{-\Delta, 0, +\Delta\}$ FOR THAT REGION. RESULTS ARE REPORTED AS ASR(%) AND AVERAGE QUERIES (AVG.Q) ON LFW (L) AND CELEBA-HQ (C)

Model	Original		End-to-End	
	ASR (L/C)	Avg.Q (L/C)	ASR (L/C)	Avg.Q (L/C)
IR50	100.0/87.0	2007/1968	89.0/90.0	1391/1431
MobileFace	100.0/89.4	1638/1359	88.0/93.0	1105/1076
CurricularFace	94.8/96.2	1216/1484	100.0/96.0	897/1127
InsightFace	100.0/91.4	1252/1553	99.0/97.0	907/1167
ShuffleFace	99.6/64.8	2161/6150	83.0/14.0	1494/2387
ArcFace	94.6/96.8	1635/2053	94.0/89.0	1165/1569
Centerloss	67.0/70.4	4869/4886	44.0/49.0	1925/1924

but decreases for others. This comparison reflects a trade-off between pipeline query reduction and attack robustness across different architectures.

VII. DISCUSSION

A. Countermeasures

We consider targeted adversarial training, artifacts detection and multi-FR model fusion as the countermeasures of PPOM-ATTACK and other adversarial attack methods.

1) *Targeted Adversarial Training*: To enhance the robustness of these FR models, adversarial training can be conducted against adversarial attacks. In the experiments targeting adversarially trained FR models, the ASR of PPOM-ATTACK and other adversarial attack methods decrease compared to that against normal FR models. Therefore, targeted adversarial training can mitigate adversarial attacks.

2) *Artifacts Detection*: It is possible to capture the artifacts of adversarial face images generated by PPOM-ATTACK and other SOTA methods, as compared to natural face images, by employing feature extraction, such as spectrum-based [46] technique, feature-hybrid-based [47] technique, etc. The anomalies can be utilized for the detection of adversarial face images, similar to deepfake detection.

3) *Multi-FR Model Fusion*: The fusion of multiple FR models enhances the robustness of the target FR model by integrating the results from various FR models, thereby providing defense against adversarial attacks. The diversity of these models and the combined features create barriers for adversaries.

B. Limitation

Since PPOM-ATTACK need to train an agent to predict perturbations with the original face image as the input and the feedback of the target FR model as the reward. If the feedback is too sparse, the efficiency of PPOM-ATTACK might decrease. Hence, the agent need to interact with the target FR model several times, which might be captured by some monitor systems and treated as abnormal behaviours. In this situation, although the adversarial face image can successfully fool the target FR models, the cybersecurity manager may suspect the “anomalies” and take actions to deny the access.

C. Future

We note that our framework can be extended with explicit adaptive perturbation scaling, e.g., dynamically shrinking or

expanding region size or offset magnitude based on real-time embedding reduction, allowing even finer control of the ASR–quality trade-off. This provides a promising direction for future work in scenarios.

VIII. ETHICS STATEMENT

To reveal the security risks inherent in FR, adversarial attacks against FR models will spur researchers to enhance the robustness of these models. Building on this, we propose PPOM-ATTACK to target several public FR models. Our experimental results show that existing FR models, including commercial ones like Face++, are not infallible and can be easily deceived.

All experiments were conducted using publicly available face image datasets. The face images used for real-world attacks were collected legally and without personal information.

To mitigate the potential real-world impacts of these attacks, this paper proposes three countermeasures (outlined in Section VII-A): targeted adversarial training, artifact detection, and multi-FR model fusion. We discuss the practical feasibility of these measures in defending against PPOM-ATTACK and other adversarial attacks. By implementing our countermeasures, we aim to raise awareness of the security risks in FR and improve the robustness of existing FR models.

Despite the threats identified in this paper, it is important to note that a real-world adversary would still find it challenging to bypass practical FR systems due to the monitoring of abnormal behavior (discussed in Section VII-B) and the continued role of cybersecurity managers in detecting anomalies.

IX. CONCLUSION

In this paper, we directly obtain feedback from the target model instead of using substitute models. We first propose a proximal policy optimization (PPO)-based agent to predict the perturbation regions in the facial image and self-adaptively disturbing the regions. Then we propose a minimum brightness offsets method specifically designed to generate perturbations that minimize the feature embedding difference between the adversarial and targeted face images. The experimental results show that our approach outperforms SOTA FR attack methods in terms of attack success rate and image quality on several FR models. In the future, we will expand our approach to speaker recognition and some multi-modal models scenes.

REFERENCES

- [1] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," 2019, *arXiv:1706.06083*.
- [2] J. Zou, Z. Pan, J. Qiu, X. Liu, T. Rui, and W. Li, "Improving the transferability of adversarial examples with resized-diverse-inputs, diversity-ensemble and region fitting," in *Proc. ECCV*, 2020, pp. 563–579.
- [3] X. Yang et al., "Towards face encryption by generating adversarial identity masks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3877–3887.
- [4] S. Hu et al., "Protecting facial privacy: Generating adversarial identity masks via style-robust makeup transfer," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 15014–15023.
- [5] F. Shamshad, M. Naseer, and K. Nandakumar, "CLIP2Protect: Protecting facial privacy using text-guided makeup via adversarial latent search," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 20595–20605.
- [6] Q. Li, Y. Hu, Y. Liu, D. Zhang, X. Jin, and Y. Chen, "Discrete point-wise attack is not enough: Generalized manifold adversarial attack for face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 20575–20584.
- [7] D. Liu, X. Wang, C. Peng, N. Wang, R. Hu, and X. Gao, "Adv-diffusion: Imperceptible adversarial face identity attack via latent diffusion model," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 4, pp. 3585–3593.
- [8] B. Yin et al., "Adv-makeup: A new imperceptible and transferable attack on face recognition," in *Proc. Thirtieth Int. Joint Conf. Artif. Intell.*, Aug. 2021, pp. 1252–1258.
- [9] J. Lin, Y. Li, and G. Yang, "FPGAN: Face de-identification method with generative adversarial networks for social robots," *Neural Netw.*, vol. 133, pp. 132–147, Jan. 2021.
- [10] Y. Zhong and W. Deng, "Towards transferable adversarial attack against deep face recognition," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 1452–1466, 2021.
- [11] Z. Wu, Y. Cheng, S. Zhang, X. Ji, and W. Xu, "UniID: Spoofing face authentication system by universal identity," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2024, pp. 1–16.
- [12] S. Kim, Y. K. Tan, B. Jeong, S. Mondal, K. M. Mi Aung, and J. Hong Seo, "Scores tell everything about bob: Non-adaptive face reconstruction on face recognition systems," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2024, pp. 1684–1702.
- [13] H. Xia, R. Zhang, Z. Kang, S. Jiang, and S. Xu, "Enhance stealthiness and transferability of adversarial attacks with class activation mapping ensemble attack," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2024, pp. 1–18.
- [14] R. Sheatsley, B. Hoak, E. Pauley, and P. McDaniel, "The space of adversarial strategies," in *Proc. 32nd USENIX Secur. Symp.*, Aug. 2022, pp. 3745–3761.
- [15] J. Prokos et al., "Squint hard enough: Attacking perceptual hashing with adversarial machine learning," in *Proc. 32nd USENIX Secur. Symp.*, Aug. 2023, pp. 211–228.
- [16] S. Shan, W. Ding, E. Wenger, H. Zheng, and B. Y. Zhao, "Post-breach recovery: Protection against white-box adversarial examples for leaked DNN models," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2022, pp. 2611–2625.
- [17] Z. Li et al., "Sibling-attack: Rethinking transferable adversarial attacks against face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 24626–24637.
- [18] D. Zheng, W. Ke, X. Li, Y. Duan, G. Yin, and F. Min, "Enhancing the transferability of adversarial attacks via multi-feature attention," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 1462–1474, 2025.
- [19] H. Ma, K. Xu, X. Jiang, Z. Zhao, and T. Sun, "Transferable black-box attack against face recognition with spatial mutable adversarial patch," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 5636–5650, 2023.
- [20] Q. Zhang, Q. Guo, R. Gao, F. Juefei-Xu, H. Yu, and W. Feng, "Adversarial relighting against face recognition," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 9145–9157, 2024.
- [21] F. Zhou, B. Yin, H. Ling, Q. Zhou, and W. Wang, "Improving the transferability of adversarial attacks on face recognition with diverse parameters augmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 3516–3527.
- [22] X. Feng, D. Ming, P. Ren, and Y. Wang, "Boosting the transferability of adversarial attack on vision transformer with adaptive token tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2024, pp. 20887–20918.
- [23] S. Jia et al., "Adv-attribute: Inconspicuous and transferable adversarial attack on face recognition," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, pp. 34136–34147.
- [24] C. Hu, Y. Li, Z. Feng, and X. Wu, "Toward transferable attack via adversarial diffusion in face recognition," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 5506–5519, 2024.
- [25] Y. Dong et al., "Efficient decision-based black-box adversarial attacks on face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7706–7714.
- [26] S. Zhang, B. Zheng, P. Jiang, L. Zhao, C. Shen, and Q. Wang, "Perception-driven imperceptible adversarial attack against decision-based black-box models," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 3164–3177, 2024.
- [27] M. F. Reza, A. Rahmati, T. Wu, and H. Dai, "CGBA: Curvature-aware geometric black-box attack," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 124–133.

- [28] F. Croce, M. Andriushchenko, N. D. Singh, N. Flammarion, and M. Hein, "Sparse-RS: A versatile framework for query-efficient sparse black-box adversarial attacks," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 6437–6445.
- [29] V. Q. Vo, E. Abbasnejad, and D. C. Ranasinghe, "BruSLeAttack: A query-efficient score-based black-box sparse adversarial attack," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024, pp. 54462–54499.
- [30] Z. Chen, B. Li, S. Wu, S. Ding, and W. Zhang, "Query-efficient decision-based black-box patch attack," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 5522–5536, 2023.
- [31] M. F. Reza, R. Jin, T. Wu, and H. Dai, "GSBAK: Top-K geometric score-based black-box attack," in *Proc. 13th Int. Conf. Learn. Represent.*, 2025, pp. 29205–29236.
- [32] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.
- [33] G. B. Huang, M. A. Mattar, T. L. Berg, and E. Learned-Miller, "Labeled faces in the wild: A database for studying face recognition in unconstrained environments," in *Proc. Workshop Faces Real-Life Images, Detection, Alignment, Recognit.*, Marseille, France: Erik Learned-Miller and Andras Ferencz and Frédéric Jurie, Oct. 2008.
- [34] T. Karras, T. Aila, S. Laine, and J. Lehtinen, "Progressive growing of gans for improved quality, stability, and variation," 2017, *arXiv:1710.10196*.
- [35] Q. Cao, L. Shen, W. Xie, O. M. Parkhi, and A. Zisserman, "VGGFace2: A dataset for recognising faces across pose and age," in *Proc. 13th IEEE Int. Conf. Autom. Face Gesture Recognit. (FG)*, May 2018, pp. 67–74.
- [36] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
- [37] S. Chen, Y. Liu, X. Gao, and Z. Han, "MobileFaceNets: Efficient CNNs for accurate real-time face verification on mobile devices," in *Proc. Chin. Conf. Biometric Recognit.*, Cham, Switzerland: Springer, 2018, pp. 428–438.
- [38] Y. Huang et al., "CurricularFace: Adaptive curriculum learning loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5900–5909.
- [39] J. Guo, J. Deng, X. An, and J. Yu. (2021). *Deepinsight/insightface: State-of-the-Art 2D and 3D Face Analysis Project*. [Online]. Available: <https://github.com/deepinsight/insightface>
- [40] Y. Martinez-Diaz, L. S. Luevano, H. Mendez-Vazquez, M. Nicolas-Diaz, L. Chang, and M. Gonzalez-Mendoza, "ShuffleFaceNet: A lightweight face architecture for efficient and highly-accurate face recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 2721–2728.
- [41] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4685–4694.
- [42] Y. Wen, K. Zhang, Z. Li, and Y. Qiao, "A discriminative feature learning approach for deep face recognition," in *Proc. Eur. Conf. Comput. Vision. (ECCV)*, Cham, Switzerland: Springer, 2016, pp. 499–515.
- [43] H. Zhang, Y. Yu, J. Jiao, E. P. Xing, L. E. Ghaoui, and M. I. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, vol. 97, pp. 7472–7482.
- [44] J. Dan et al., "TransFace: Calibrating transformer training for face recognition from a data-centric perspective," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 20585–20596.
- [45] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," *IEEE Signal Process. Lett.*, vol. 23, no. 10, pp. 1499–1503, Oct. 2016.
- [46] P. Harder, F.-J. Pfreundt, M. Keuper, and J. Keuper, "SpectralDefense: Detecting adversarial attacks on CNNs in the Fourier domain," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2021, pp. 1–8.
- [47] J. Zhang, K. Cheng, G. Sovernigo, and X. Lin, "A heterogeneous feature ensemble learning based deepfake detection method," in *Proc. IEEE Int. Conf. Commun.*, May 2022, pp. 2084–2089.



Ke Cheng received the B.S. degree from Hubei University of Technology, Wuhan, China, in 2023. He is currently pursuing the M.S. degree with the School of Hubei University of Technology. In 2022, he worked as an Intern at Topsec Technologies, where he focused on vulnerability mining. His research interests are security issues in artificial intelligence and artificial intelligence-aided security.



Jixin Zhang (Member, IEEE) received the M.S. degree in computer science and technology from Wuhan University of Technology in 2014 and the Ph.D. degree in computer science and technology from Hunan University in 2019. After that, he visited and studied at The Chinese University of Hong Kong for one year. He is currently an Associated Professor with the School of Computer Science, Hubei University of Technology. His active research interests include artificial intelligence and security.



Haiyun Li received the B.S. degree from the Huazhong University of Science and Technology in 2019 and the M.S. degree in electronic information from Wuhan University of Technology, Wuhan, China, in 2023. He is currently pursuing the Ph.D. degree in electronic information with Tsinghua University. His research interests include artificial intelligence security, speech, and language processing.



Zipeng Zhong received the B.S. degree from Hubei University of Technology, Hubei, China, in 2023. From 2024 to 2025, he was an Artificial Intelligence Algorithm Engineer at Guangzhou TEC Solutions Company Ltd. After resigning in 2025, he joined Hubei University of Technology as an Assistant Researcher. His research interests are security issues in artificial intelligence and artificial intelligence-aided security.



Mingwu Zhang was a JSPS Post-Doctoral Fellow of Japan Society for the Promotion of Sciences at the Institute of Mathematics for Industry, Kyushu University, Japan, from 2010 to 2012. From 2015 to 2016, he was a Senior Visiting Scholar at the School of Computing and Information Technology, University of Wollongong, Australia. He is currently a Professor at the School of Computer Science, Hubei University of Technology, China. His research interests include cryptography, privacy, and security.



Zheng Qin (Member, IEEE) received the Ph.D. degree in computer software and theory from Chongqing University in 2001. From 2010 to 2011, he was a Visiting Scholar with the Department of Computer Science, Michigan University. He is currently a Professor with the College of Computer Science and Electronic Engineering, Hunan University, where he is the Vice Dean. His research interests include privacy and security.